# Supplementary Material

## Fisher-Glass Learning: Classification Bounds and Subpopulation Audits

**El Mustapha Mansouri**
School of Engineering, Institute of Science Tokyo
Yokohama, Kanagawa 226-8501, Japan
mansouri.e.2224@m.isct.ac.jp

## Guide to the Supplemental Material

This supplement collects the proofs, estimation details, experimental protocols, and additional results supporting *Fisher-Glass Learning: Classification Bounds and Subpopulation Audits*. Appendix A gives the theoretical arguments, including the worked classification family. Appendices B and C define the operational matrices, uncertainty calculations, and frozen-gate protocol. Appendices D and E report the empirical checks and information-changing interventions. The map below locates the completed comparisons and their principal interpretation limits. Appendix letters and result numbering A–E match the accompanying paper. References to main-text sections and results link to `main.pdf`; appendix references are internal. Appendix F gives the evidence and replay index.

| Comparison | Protocol and result to read together | Location |
|---|---|---|
| Frozen background intervention | Validation selects the waterbird/water group before held-out scoring. Same-type and opposite-type recompositions share bird foregrounds and labels. The controlled accuracy difference is 52.6 percentage points across five training seeds; confident-flip results are reported separately. | Appendix D.2; Table D.3 |
| Selection and summary controls | Accuracy and negative log-likelihood (NLL) selectors agree with the surviving-fraction rule on the same eligible set. Centering reverses the selection; the specified predictive-Fisher construction preserves it. The four-summary comparison does not establish superior swap-damage prediction by inverse-CVaR. | Tables D.4–D.5, D.6 |
| Auxiliary failure prediction | Species-held-out prediction adds background and residual coordinates to source loss, confidence, and logit predictors. Gains within the selected group are positive across seeds; pooled average precision decreases when group indicators are included. This does not isolate a scalar-information or residual-only effect. | Appendix D.2; Tables D.7, F.2 |
| Ridge and exact-information audits | The 75 default-ridge audits report nuisance conditioning, ridge inflation, exact plug-in values, and nominal conditional Wald lower values. Fresh training audits are separate from historical training cohorts; numerical lower/exact ordering alone does not establish confidence coverage. | Appendix B.4; Tables B.1, B.2 |

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

| Comparison | Protocol and result to read together | Location |
|---|---|---|
| Hidden-slice discovery | Task labels and a fixed nuisance direction define the search; group annotations remain sealed through evaluation. Matched-mass selectors, random-direction and feature-moment controls, support, stability, and post hoc composition are reported. CLIP versus random direction remains inconclusive. | Appendices C, D.4; Table D.9 |
| Constrained slice repair | Loss, Fisher, and combined repairs share accuracy constraints and an unchanged candidate. Fisher repair selects that unchanged model in all five seeds. Group-supervised Group DRO improves worst-group accuracy without raising the rediscovered slice's mean information. | Appendix E.3; Table E.3 |
| Group-free coefficient selection | This separate post hoc audit selects the same coefficients as validation-NLL minimization. Information increases while worst-group accuracy decreases by 12.6 points on average. The proposed threshold-and-risk-budget policy is distinct and has not been prospectively evaluated. | Appendix E.3; Table E.4 |

**Cohorts and scope.** The primary prospective intervention uses seeds 22–26 and the surviving-fraction rule. The earlier development cohort uses seeds 17–21 and an absolute-information rule; both use the same benchmark. Matched selectors, matrix constructions, and the four-summary comparison analyze retained records after the intervention outcomes were known. Hidden-slice discovery and its constrained repair use the corrected v2 cohort, distinct from the post hoc group-free coefficient cohort. No retained experiment tests whether the learned low-information slice is more sensitive than a confidence-matched population. Intervals across seeds describe fitting or partition variation on fixed data, not independent datasets.

**Provenance and replay.** The accompanying artifact's `START_HERE.md` and `docs/MANUSCRIPT_MAP.md` map these results to retained inputs, hashes, and replay commands. Numerical replay uses saved records and does not retrain the image classifiers. Historical synthetic, diagnostic, training, and acquisition summaries have explicitly documented gaps in their original per-seed outputs; fresh audits and regenerated figures do not replace those missing records. Acquisition means therefore carry no reconstructed historical intervals. The experiment guides distinguish independent replication from exact replay.

# A Proofs of the Core Results

Throughout, task coordinates use the common normalization of Section 2, and environments follow the declared distribution $\Pi$.

## A.1 Projection, Normalization, and Task–Nuisance Tangency

Use the centered, square-integrable scores and blocks $J_{AB,e}$ of Section 2.1. Suppose the $d_N$ coordinates of the finite nuisance-score vector span $\mathcal{T}_{N,e}$ and $J_{NN,e} \succ 0$. With $\Gamma_e = J_{TN,e}J_{NN,e}^{-1}$,

$$\begin{aligned} \mathbb{E}_{P_e}[(s_{T,e} - \Gamma_e s_{N,e})s_{N,e}^\top] &= J_{TN,e} - \Gamma_e J_{NN,e} = 0, \\ F_e = \mathbb{E}_{P_e}[s_{T,e}^{\text{eff}}(s_{T,e}^{\text{eff}})^\top] &= J_{TT,e} - J_{TN,e}J_{NN,e}^{-1}J_{NT,e}. \end{aligned} \tag{A.1}$$

Because $\Gamma_e s_{N,e}$ lies componentwise in the nuisance span, orthogonality identifies the residual as the efficient score in Equation (1) [4, 27]. Equivalently,

$$u^\top F_e u = \min_{v \in \mathbb{R}^{d_N}} \mathbb{E}_{P_e}[(u^\top s_{T,e} - v^\top s_{N,e})^2], \qquad v^\star = J_{NN,e}^{-1}J_{NT,e}u. \tag{A.2}$$

Hence $F_e \succeq 0$, with $u^\top F_e u = 0$ exactly when $u^\top s_{T,e}$ belongs to the nuisance span. The projection definition extends directly to the closed span of an infinite-dimensional nuisance tangent space.

For separate invertible transformations $\widetilde{s}_T = Us_T$ and $\widetilde{s}_N = Vs_N$, the nuisance span is unchanged and

$$\widetilde{s}_T^{\text{eff}} = Us_T^{\text{eff}}, \qquad \widetilde{F} = UFU^\top; \qquad \frac{F}{J_{TT}} = 1 - \rho^2 \quad \text{in the scalar case.} \tag{A.3}$$

Here $\rho = J_{TN}/\sqrt{J_{TT}J_{NN}}$ requires positive raw task and nuisance information. Exact information is invariant to nuisance-basis changes; task congruence preserves rank but generally changes eigenvalues. The scalar surviving fraction is invariant to nonzero task rescaling and invertible nuisance-basis changes. Nuisance ridge preserves the latter invariance only if $\lambda_{\text{rid}}I$ becomes $\lambda_{\text{rid}}VV^\top$ under $s_N \mapsto Vs_N$ (Appendix B).

For the tangency construction in Section 2.2, let $b_e, r_e \in L_0^2(P_e)$ be orthogonal unit vectors, $\mathcal{T}_{N,e} = \text{span}\{b_e\}$, and $a_e = b_e + \delta_e r_e$ with $\delta_e \geq 0$. The projection coefficient is $\langle a_e, b_e\rangle_e/\|b_e\|_e^2 = 1$, so

$$a_e^{\text{eff}} = \delta_e r_e, \qquad F_e = \|\delta_e r_e\|_e^2 = \delta_e^2, \qquad J_{TT,e} = 1 + \delta_e^2, \qquad \frac{F_e}{J_{TT,e}} = \frac{\delta_e^2}{1+\delta_e^2}. \tag{A.4}$$

This establishes Equation (3); coexistence with low prediction risk additionally requires a successful nuisance-based predictor.

## A.2 Normal Coordinates and the Inverse-Information Tail

The full assumptions of Theorem 2 are as follows. The zero set $\mathcal{Z} = \{\lambda = 0\}$ is a nonempty compact $C^2$ embedded manifold without boundary, of codimension $c \in \{1, \ldots, d\}$. Write $U_r(\mathcal{Z}) = \{z : \text{dist}(z, \mathcal{Z}) < r\}$ for its tube, $N_s\mathcal{Z}$ for the normal space at $s$, $S(N_s\mathcal{Z})$ for its unit sphere, and $\text{vol}_{\mathcal{Z}}$ for induced volume. The nonnegative $\lambda$ is continuous on $\overline{U}_{r_\star}(\mathcal{Z})$ and bounded away from zero outside it. In coordinates $z = s + r\omega$, assume uniformly over $\omega \in S(N_s\mathcal{Z})$ and $s \in \mathcal{Z}$

$$\lambda(s + r\omega) = a(s,\omega)r^{2m}(1 + o(1)), \qquad p(s + r\omega) = b(s,\omega)r^\nu(1 + o(1)), \tag{A.5}$$

where $p$ is the Lebesgue density, $m > 0$, $\nu > -c$, and continuous $a, b$ have positive uniform lower and upper bounds. Put

$$\beta = \frac{c+\nu}{2m}, \qquad C_{\mathcal{Z}} = \frac{1}{c+\nu}\int_{\mathcal{Z}}\int_{S(N_s\mathcal{Z})} b(s,\omega)a(s,\omega)^{-\beta}\, d\sigma_s(\omega)\, d\,\text{vol}_{\mathcal{Z}}(s). \tag{A.6}$$

The sphere measure $\sigma_s$ is unnormalized; for $c = 1$, each of the two normal directions has mass one.

*Proof of Theorem 2.* For any $0 < r_0 < r_\star$, continuity and absence of zeros on the compact annulus $r_0 \leq \text{dist}(z, \mathcal{Z}) \leq r_\star$ give a positive minimum of $\lambda$. Together with the bound outside the outer tube, this localizes $\{\lambda < t\}$ to $U_{r_0}(\mathcal{Z})$ for all sufficiently small $t$. The tubular volume element is

$dz = j(s, r\omega) r^{c-1}\, dr\, d\sigma_s\, d\,\mathrm{vol}_{\mathcal{Z}}$, where the coordinate Jacobian satisfies $j = 1 + O(r)$ uniformly. Thus, for any $0 < \zeta < 1$, uniformity permits $r_0$ small enough that

$$\begin{aligned} (1-\zeta)ar^{2m} \le \lambda(s + r\omega) \le (1+\zeta)ar^{2m}, \\ (1-\zeta)br^{\nu} \le p(s + r\omega) j(s, r\omega) \le (1+\zeta)br^{\nu} \end{aligned} \qquad (0 < r < r_0). \tag{A.7}$$

For small $t$, all radial thresholds lie inside this tube. The event contains $r < (t/((1+\zeta)a))^{1/(2m)}$ and is contained in $r < (t/((1-\zeta)a))^{1/(2m)}$. Integrating uses $\int_0^{(t/a)^{1/(2m)}} br^{c+\nu-1}\, dr = ba^{-\beta}t^{\beta}/(c+\nu)$ and gives

$$(1-\zeta)(1+\zeta)^{-\beta} C_{\mathcal{Z}} \le \frac{\Pr(\lambda < t)}{t^{\beta}} \le (1+\zeta)(1-\zeta)^{-\beta} C_{\mathcal{Z}}. \tag{A.8}$$

Let $t \downarrow 0$ and then $\zeta \downarrow 0$. Compactness and the coefficient bounds ensure $0 < C_{\mathcal{Z}} < \infty$, so $\Pr(\lambda < t) \sim C_{\mathcal{Z}} t^{\beta}$. Since $c \ge 1$, $\mathcal{Z}$ has zero Lebesgue and environment measure; substituting $t = x^{-1}$ proves the inverse tail in the next display. □

The tail-integral identity then yields, for every $p_0 > 0$,

$$\begin{aligned} &\Pr(\lambda^{-1} > x) \sim C_{\mathcal{Z}} x^{-\beta}, \\ &\mathbb{E}[\lambda^{-p_0}] = p_0 \int_0^{\infty} x^{p_0 - 1} \Pr(\lambda^{-1} > x)\, dx < \infty \quad \Longleftrightarrow \quad p_0 < \beta. \end{aligned} \tag{A.9}$$

The bounded-$x$ part is finite; the tail has the convergence criterion of $\int^{\infty} x^{p_0 - \beta - 1}\, dx$. In particular, the inverse mean is finite exactly for $\beta > 1$, with logarithmic divergence at $\beta = 1$.

**Scalar branches.** For $\lambda = \delta^2$, $\Pr(\delta < t) \sim C_{\delta} t^{\gamma}$ gives exponent $\gamma/2$. More generally, let $\lambda_K = \sum_{k=1}^{K} w_k \delta_k^2$, $w_k > 0$, where $\delta \in [0, \infty)^K$ has joint density $f$ continuous and positive at the origin from within the orthant. Changing variables gives the scalar portfolio law [13]:

$$\begin{aligned} \Pr(\lambda_K < t) &= \frac{t^{K/2}}{\sqrt{\prod_k w_k}} \int_{B_K^+} f\big(\sqrt{t/w_1}\, u_1, \ldots, \sqrt{t/w_K}\, u_K\big)\, du \\ &\sim \frac{f(0) v_K}{2^K \sqrt{\prod_k w_k}}\, t^{K/2}. \end{aligned} \tag{A.10}$$

Here $B_K^+$ is the positive-orthant unit ball and $v_K$ the full ball's volume. Independence gives $f(0) = \prod_k f_k(0+)$ for branch marginal densities $f_k$; the inverse mean is finite for $K \ge 3$. This calculation concerns additive scalar information or a fixed task direction. A minimum-eigenvalue conclusion instead requires applying Theorem 2 to that eigenvalue.

### A.3 Local Classification: Testing Reduction and Tail Aggregation

Let $\mathcal{E}_P(g) = R_P(g) - R_P^{\star}$. Fix a local family $\mathcal{P}_e^{\mathrm{loc}}$ independently of sample size, and define

$$\mathfrak{R}_e(n) = \inf_{\widehat{g}_n} \sup_{P \in \mathcal{P}_e^{\mathrm{loc}}} \mathbb{E}_{S_n \sim P^{\otimes n}}[\mathcal{E}_P(\widehat{g}_n(S_n))], \tag{A.11}$$

where learning rules are measurable and $n$ counts independent observations from environment $e$; independent learner randomization is included in the expectation. Assume $\lambda_e > 0$ and a unit weakest efficient-information direction $u_e$. For every $0 < t \le t_{0,e}$, the fixed family contains $P_{e,\pm t}$ with task coordinates $\pm t u_e$ and potentially varying nuisance coordinates. For fixed $K_e, c_e, t_{0,e} > 0$ and $q > 0$, assume both bounds in Equation (4) hold throughout that radius.

*Proof of Theorem 1.* Fix an admissible $t$ and abbreviate $P_{\pm} = P_{e,\pm t}$. For any learning rule, the decision-separation condition $\mathcal{E}_{P_+}(g) + \mathcal{E}_{P_-}(g) \ge c_e t^q$ implies that choosing the alternative with smaller excess risk makes any testing error cost at least $c_e t^q/2$ under the true alternative. Explicitly,

$$\begin{aligned} \psi_t(S_n) = + \iff \mathcal{E}_{P_+}(\widehat{g}_n(S_n)) \le \mathcal{E}_{P_-}(\widehat{g}_n(S_n)), \\ \max_{\sigma \in \{+,-\}} \mathbb{E}_{P_\sigma^{\otimes n}}[\mathcal{E}_{P_\sigma}(\widehat{g}_n)] \ge \frac{c_e t^q}{4} \{P_+^{\otimes n}(\psi_t = -) + P_-^{\otimes n}(\psi_t = +)\}. \end{aligned} \tag{A.12}$$

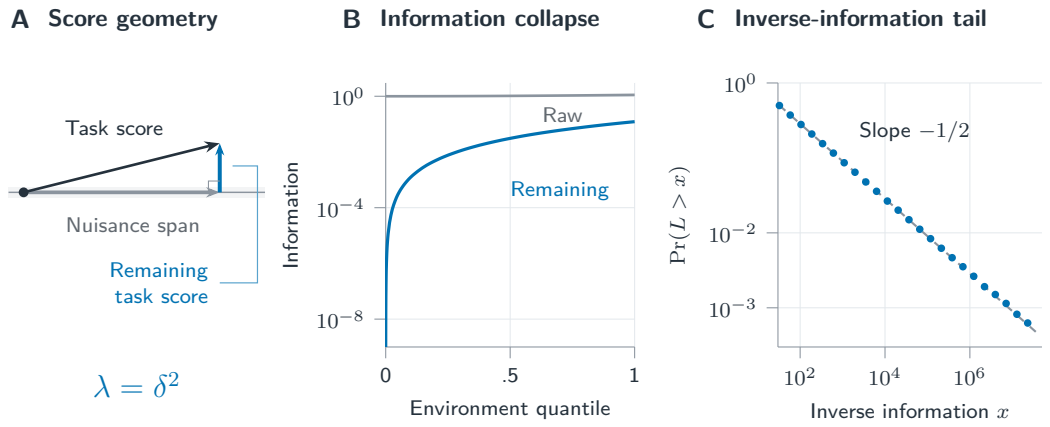



Figure A.1: **Near-alignment produces a heavy inverse-information tail.** (A) Projection leaves a residual score of norm $\delta$, giving information $\lambda = \delta^2$. (B) Raw information $1 + \delta^2$ remains near one as $\lambda$ approaches zero. (C) Empirical survival probabilities (blue) follow the $x^{-1/2}$ reference (dashed). Panels B–C use 300,000 environments with $\delta \sim \mathrm{Uniform}(0, 0.35)$ and $L = (\delta^2 + 10^{-12})^{-1}$.

The standard exponential two-point testing inequality [28, Lemma 2.6], together with product additivity of KL and the assumed KL bound, gives

$$\begin{aligned} P_+^{\otimes n}(\psi_t = -) + P_-^{\otimes n}(\psi_t = +) &\geq \tfrac{1}{2}\exp[-n\,\mathrm{KL}(P_+\|P_-)] \geq \tfrac{1}{2}e^{-nK_e t^2 \lambda_e}, \\ \mathfrak{R}_e(n) &\geq \frac{c_e}{8} t^q e^{-nK_e t^2 \lambda_e}. \end{aligned} \tag{A.13}$$

The second line takes the supremum over the fixed family containing both alternatives and then the infimum over learning rules. Choose

$$\begin{aligned} t_n &= \min\{t_{0,e}, (nK_e\lambda_e)^{-1/2}\}, \qquad nK_e\lambda_e t_n^2 \leq 1, \\ \mathfrak{R}_e(n) &\geq c_0 c_e \min\{t_{0,e}^q, (nK_e\lambda_e)^{-q/2}\}, \qquad c_0 = \frac{e^{-1}}{8}. \end{aligned} \tag{A.14}$$

Independent learner randomization has the same law under both alternatives and leaves their KL divergence unchanged, so the argument also covers randomized rules. □

**Quadratic decision separation.** Suppose the binary alternatives share a feature marginal $Q_e$, with regression functions $\eta_{\pm,t}(x) = P_{e,\pm t}(Y = 1 \mid X = x)$ and Bayes rules $g^\star_{\pm,t}(x) = \mathbf{1}\{\eta_{\pm,t}(x) \geq 1/2\}$. Let $D_{e,t} = \{g^\star_{+,t} \neq g^\star_{-,t}\}$ be their disagreement set. Applying the binary excess-risk identity [3, Eq. (1.1)] gives

$$\begin{aligned} &\inf_g \{\mathcal{E}_{P_{e,+t}}(g) + \mathcal{E}_{P_{e,-t}}(g)\} \\ &\qquad = \int_{D_{e,t}} \min\{|2\eta_{+,t}(x) - 1|, |2\eta_{-,t}(x) - 1|\}\, dQ_e(x). \end{aligned} \tag{A.15}$$

Pointwise, a classifier must disagree with at least one Bayes rule on $D_{e,t}$; choosing the smaller margin attains the integral for unrestricted measurable binary classifiers, while it remains a lower bound for a restricted class. If, for every $0 < t \leq t_{0,e}$, a measurable $A_{e,t} \subseteq D_{e,t}$ has $Q_e(A_{e,t}) \geq c_{\mathrm{mass},e}t$ and both margins at least $c_{\mathrm{margin},e}t$, with positive constants independent of $t$, the integral is at least $c_{\mathrm{mass},e}c_{\mathrm{margin},e}t^2$. Thus $q = 2$ and $c_e = c_{\mathrm{mass},e}c_{\mathrm{margin},e}$ satisfy decision separation; the KL condition is still required separately.

**Environmental aggregation.** If these assumptions hold for $\Pi$-almost every environment and all quantities are measurable, monotonicity and positive homogeneity of CVaR give

$$\begin{aligned} \mathcal{B}_e(n) &= c_e \min\{t_{0,e}^q, (nK_e\lambda_e)^{-q/2}\}, \\ \mathrm{CVaR}_{\alpha,e\sim\Pi}(\mathfrak{R}_e(n)) &\geq c_0\, \mathrm{CVaR}_{\alpha,e\sim\Pi}(\mathcal{B}_e(n)). \end{aligned} \tag{A.16}$$

Separation $c_e$, distinguishability $K_e$, and radius $t_{0,e}$ remain inside the aggregation. For $q = 2$ and $nK_e\lambda_e \geq t_{0,e}^{-2}$, $\mathcal{B}_e(n) = c_e/(nK_e\lambda_e)$; outside that interior regime the radius truncates the bound.

### A.4 A Boundary-Region Classification Family

Fix a known $0 < \delta \leq 1/2$ and let $\mathcal{P}_\delta$ be the family in Equation (6), with $X \sim \mathrm{Unif}[-1, 1]$ and parameter rectangle $|\theta|, |\mu| \leq \delta/8$. Define

$$\mathfrak{R}_\delta(n) = \inf_{\widehat{g}_n} \sup_{|\theta|,|\mu|\leq\delta/8} \mathbb{E}_{\theta,\mu}[R_{\theta,\mu}(\widehat{g}_n) - R^\star_{\theta,\mu}],$$

where the infimum ranges over measurable, possibly randomized learning rules based on $n$ independent observations.

**Proposition A.1** (Nuisance confounding near a classification boundary)**.** *At the reference parameter* $(0, 0)$*, the raw and efficient task information satisfy*

$$J_{TT} = 4\log 3, \qquad F_\delta = 4\log\frac{2+\delta}{2-\delta}, \qquad 4\delta \leq F_\delta \leq \frac{64}{15}\delta.$$

*For every* $n \geq 1$*,*

$$\frac{1}{128e}\min\{\delta^2, (n\delta)^{-1}\} \leq \mathfrak{R}_\delta(n) \leq \min\left\{\frac{\delta^2}{32}, \frac{1-(1-\delta)^{n+1}}{2(n+1)\delta}\right\}. \tag{A.17}$$

*Consequently, the rate in Equation* (8) *holds with constants independent of* $n$ *and* $\delta$*.*

*Proof.* Write $b_\delta(x) = \mathbf{1}\{|x| > \delta\}$ and $\eta_0(x) = 1/2 + x/4$. Throughout the rectangle, $1/8 \leq \eta_{\theta,\mu} \leq 7/8$, so the binary likelihood is regular in $(\theta, \mu)$ for each fixed $\delta > 0$. Inside $|x| \leq \delta$, the regression crosses $1/2$ at $-4\theta \in [-\delta/2, \delta/2]$. Outside, $|\theta + \mu| \leq \delta/4 < |x|/4$, so its sign relative to $1/2$ is the sign of $x$. Thus

$$g^\star_{\theta,\mu}(x) = \mathbf{1}\{x \geq -4\theta\}, \qquad \mathcal{E}_{\theta,\mu}(g_a) = 2(a-\theta)^2 \quad \text{for } g_a(x) = \mathbf{1}\{x \geq -4a\},\ |a| \leq \delta/8. \tag{A.18}$$

The second identity integrates the margin over the interval between the two thresholds, which lies entirely inside $[-\delta, \delta]$.

**Information.** Let $v_0(x) = \eta_0(x)(1-\eta_0(x)) = (4-x^2)/16$. The reference scores are

$$s_T = \frac{Y - \eta_0(X)}{v_0(X)}, \qquad s_N = b_\delta(X)s_T.$$

Since $b_\delta^2 = b_\delta$, $J_{TN} = J_{NN} > 0$, so the nuisance-projection coefficient is one and $s_T^{\mathrm{eff}} = \mathbf{1}\{|X| \leq \delta\}s_T$. Hence

$$J_{TT} = \int_{-1}^{1}\frac{8}{4-x^2}\,dx = 4\log 3, \qquad F_\delta = \int_{-\delta}^{\delta}\frac{8}{4-x^2}\,dx = 4\log\frac{2+\delta}{2-\delta}. \tag{A.19}$$

On $[-\delta, \delta]$, $2 \leq 8/(4-x^2) \leq 32/15$, giving the asserted bounds on $F_\delta$.

**Decision separation and distinguishability.** For $0 < t \leq \delta/8$, take $P_{+t} = P_{t,-t}$ and $P_{-t} = P_{-t,t}$. Their regression functions are $\eta_0 \pm t\mathbf{1}\{|x| \leq \delta\}$ and their Bayes rules disagree on $(-4t, 4t)$. The binary excess-risk identity therefore gives the exact separation

$$\inf_g\{\mathcal{E}_{+t}(g) + \mathcal{E}_{-t}(g)\} = \frac{1}{2}\int_{-4t}^{4t}\left(2t - \frac{|x|}{2}\right)dx = 4t^2. \tag{A.20}$$

For Bernoulli parameters $p, q$, $\mathrm{KL}(\mathrm{Ber}(p)\|\,\mathrm{Ber}(q)) \leq (p-q)^2/[q(1-q)]$. Here the alternatives agree outside the central region; inside, $p - q = 2t$ and $q \in [5/16, 11/16]$. Since $P(|X| \leq \delta) = \delta$ and $F_\delta \geq 4\delta$,

$$\mathrm{KL}(P_{+t}\|P_{-t}) \leq \frac{1024}{55}\delta t^2 \leq \frac{256}{55}F_\delta t^2 < 5F_\delta t^2. \tag{A.21}$$

Theorem 1 thus applies with $q = 2$, $c_e = 4$, $K_e = 5$, $t_{0,e} = \delta/8$, and $\lambda_e = F_\delta$. Its proof gives

$$\mathfrak{R}_\delta(n) \geq \frac{1}{2e}\min\left\{\frac{\delta^2}{64}, \frac{1}{5nF_\delta}\right\} \geq \frac{1}{128e}\min\{\delta^2, (n\delta)^{-1}\},$$

using $F_\delta \leq 64\delta/15$ in the last step.

**Matching upper bound, including empty central samples.** Let $I_i = \mathbf{1}\{|X_i| \le \delta\}$, $N = \sum_i I_i \sim \mathrm{Bin}(n, \delta)$, and define

$$\widehat{\theta} = \mathrm{clip}_{[-\delta/8, \delta/8]}\left(\frac{\sum_{i=1}^{n} I_i(Y_i - 1/2 - X_i/4)}{N+1}\right). \tag{A.22}$$

This estimator is zero when $N = 0$. Conditional on the feature observations, each selected residual has mean $\theta$, variance at most $1/4$, and independent noise. Before clipping, its mean-squared error is at most $(N/4 + \theta^2)/(N+1)^2 \le 1/[4(N+1)]$; clipping cannot increase squared error because the interval contains $\theta$. The identity

$$\mathbb{E}\frac{1}{N+1} = \frac{1-(1-\delta)^{n+1}}{(n+1)\delta}$$

and Equation (A.18) give the second upper bound in Equation (A.17). The fixed rule $g_0$ gives the first, since $2\theta^2 \le \delta^2/32$. Taking the better of these two known rules proves the stated upper bound. □

Only the central observations distinguish the least-favorable task alternatives: their expected number is $n\delta$. The inverse-information rate is active once $n\delta^3$ is sufficiently large; otherwise the local decision scale $\delta^2$ caps the risk. The nuisance parameter here changes conditional probabilities outside the boundary region while preserving the Bayes rule at fixed $\theta$. This is a statistical learning mechanism, not a mathematical model of the image-background intervention.

### A.5 CVaR, Information Coverage, and Stabilization

For $0 < \alpha < 1$ and nonnegative $Z$, the CVaR minimization formula holds for integrable $Z$ [21, Theorem 10]; the threshold-weight argument below gives its dual:

$$\begin{aligned} \mathrm{CVaR}_\alpha(Z) &= \inf_{\tau \in \mathbb{R}} \left\{ \tau + \frac{\mathbb{E}[(Z-\tau)_+]}{1-\alpha} \right\} = \sup_{\xi \in \mathcal{W}_\alpha} \mathbb{E}[\xi Z], \\ \mathcal{W}_\alpha &= \{\xi : 0 \le \xi \le (1-\alpha)^{-1},\ \mathbb{E}\xi = 1\}. \end{aligned} \tag{A.23}$$

Both values are $+\infty$ if $\mathbb{E}Z = +\infty$. For integrable $Z$, a maximizing weight equals $(1-\alpha)^{-1}$ above $v_\alpha = \inf\{v : \Pr(Z \le v) \ge \alpha\}$, zero below it, and fills the remaining mass at the threshold. Such fractional weighting is possible because $\Pr(Z > v_\alpha) \le 1 - \alpha \le \Pr(Z \ge v_\alpha)$. The dual gives monotonicity, positive homogeneity, and $v_\alpha \le \mathrm{CVaR}_\alpha(Z)$. For the finite audit distribution $\Pi_G$, let $\pi_g = \Pi_G(\{g\}) \ge 0$, $\sum_g \pi_g = 1$. Then

$$\mathrm{CVaR}_{\alpha,\pi}(Z_g) = \max_{\substack{\sum_g \omega_g = 1 \\ 0 \le \omega_g \le \pi_g/(1-\alpha)}} \sum_g \omega_g Z_g. \tag{A.24}$$

These normalized tail weights, including threshold atoms, also govern Appendix E.4.

*Proof of Proposition 1.* Let $\varepsilon > 0$ and $\lambda_e \in [0, \infty)$ be finite almost surely. Write $L_e^{(\varepsilon)} = (\lambda_e + \varepsilon)^{-1}$ and $I_\alpha^{\mathrm{FG}}(\varepsilon)$ for the certificate in Equation (10). If $I_\alpha^{\mathrm{FG}}(\varepsilon) \ge \rho > \varepsilon$, then

$$\begin{aligned} v_\alpha(L^{(\varepsilon)}) \le \mathrm{CVaR}_\alpha(L^{(\varepsilon)}) = [I_\alpha^{\mathrm{FG}}(\varepsilon)]^{-1} \le \rho^{-1}, \\ \Pr_{e \sim \Pi}(\lambda_e \ge \rho - \varepsilon) = \Pr_{e \sim \Pi}(L_e^{(\varepsilon)} \le \rho^{-1}) \ge \alpha. \end{aligned} \tag{A.25}$$

The last step is the quantile definition and includes equality at the information threshold. □

Since $0 < L^{(\varepsilon)} \le \varepsilon^{-1}$, always $I_\alpha^{\mathrm{FG}}(\varepsilon) \ge \varepsilon$; only $\rho > \varepsilon$ certifies a strictly positive floor. With four equally weighted groups and $\alpha = 0.95$, one group already contains the entire upper-tail mass, so

$$\mathrm{CVaR}_{0.95}(L^{(\varepsilon)}) = \max_g L_g^{(\varepsilon)}, \qquad I_{0.95}^{\mathrm{FG}}(\varepsilon) = \min_g \lambda_g + \varepsilon. \tag{A.26}$$

**Exact limit and positivity.** Set $L^{(0)} = \lambda^{-1}$, with $1/0 = +\infty$. As $\varepsilon \downarrow 0$, $L^{(\varepsilon)} \uparrow L^{(0)}$, and monotone convergence applies to $\mathbb{E}[\xi L^{(\varepsilon)}]$ for every feasible $\xi$. The supremum over $\xi$ commutes with the increasing supremum along any sequence $\varepsilon_k \downarrow 0$, giving

$$\begin{aligned} &\mathrm{CVaR}_\alpha(L^{(\varepsilon)}) \uparrow \mathrm{CVaR}_\alpha(\lambda^{-1}), \\ &\qquad I_\alpha^{\mathrm{FG}}(\varepsilon) \downarrow I_\alpha^{\mathrm{FG}}(0) := \left[\mathrm{CVaR}_\alpha(\lambda^{-1})\right]^{-1}, \qquad 1/(+\infty) = 0. \end{aligned} \tag{A.27}$$

For nonnegative $Z$, the feasible weight $\xi = 1$ and variational choice $\tau = 0$ imply

$$\mathbb{E}Z \le \mathrm{CVaR}_\alpha(Z) \le \frac{\mathbb{E}Z}{1-\alpha}, \qquad I_\alpha^{\mathrm{FG}}(0) > 0 \iff \mathbb{E}[\lambda^{-1}] < \infty. \tag{A.28}$$

Under the geometric tail assumptions, this is equivalent to $\beta = (c+\nu)/(2m) > 1$; the exact certificate vanishes for $\beta \le 1$. Information coverage is a population statement under $\Pi$, separate from the confidence level of an estimated lower certificate.

# B Estimation and Certification

The bounds condition on a predictor, coordinates, calibration, and slices fixed independently of evaluation data. Evaluation-fitted normalization requires additional error control. Implementation details and numerical records appear in the artifact's `docs/extended_appendix_details/` (see the artifact availability statement).

## B.1 Operational Models and Matrix Targets

For evaluation distribution $P_g$ and calibrated $K$-class logits $\ell_w$, maps $\Phi_T(x) \in \mathbb{R}^{K \times d_T}$ and $\Phi_N(x) \in \mathbb{R}^{K \times d_N}$ convert task and nuisance parameters into logit changes:

$$\ell_{\theta,\mu}(x) = \ell_w(x) + \Phi_T(x)\theta + \Phi_N(x)\mu, \qquad D = (\Phi_T\ \Phi_N), \quad d = d_T + d_N. \tag{B.1}$$

With $p = \mathrm{softmax}(\ell_w)$, $H = \mathrm{diag}(p) - pp^\top$, one-hot label vector $\mathbf{e}_Y$, and normalized feature coordinates $v = (v_T^\top\ v_N^\top)^\top$, distinguish

$$\begin{aligned} &M^{\mathrm{pred}} = D^\top H D, \qquad M^{\mathrm{obs}} = ss^\top, \quad s = D^\top(\mathbf{e}_Y - p), \\ &M^{\mathrm{feat}} = vv^\top, \qquad J_g = \mathbb{E}_{P_g}[M]. \end{aligned} \tag{B.2}$$

For a binary scalar logit $\ell_{\theta,\mu} = \ell_w + \theta z + \mu b$, this gives $M^{\mathrm{pred}} = p(1-p)vv^\top$, with $p = \sigma(\ell_w)$ and $v = (z, b)^\top$. Predictive scores have conditional mean zero under the fitted label distribution even when features do not; their Schur complement is efficient information for this operational model. Observed-label moments equal predictive Fisher under conditional correct specification, $P_g(Y = \cdot \mid X) = p(X)$; otherwise their centered covariance subtracts $\mathbb{E}[s]\mathbb{E}[s]^\top$. Feature geometry satisfies $J_g = \Sigma_g + \mu_g\mu_g^\top$, with $\mu_g = \mathbb{E}[v]$ and $\Sigma_g = \mathrm{Cov}(v)$. The prospective centered comparison uses within-slice covariance $\Sigma_g$ with divisor $n_g$ and unchanged global scales, removing the mean contribution without rescaling.

Prospective selection uses uncentered feature–Schur fractions; Appendix D.2 compares centered and predictive targets. Within-group normalization of $p(1-p)$ weights changes the effective nuisance ridge and differs from the predictive moment. Hidden discovery leaves this factor unnormalized and shrinks toward the same-class pooled matrix, with $\eta = 0.10$. Removing ridge leaves the declared matrix, including shrinkage, unchanged. Normalization conventions and nuisance-specification checks appear in Appendices D and D.5.

**Operational subspaces and cost.** A task coordinate specifies the predictive contrast to be audited; nuisance coordinates specify variations expected to preserve it. Candidate constructions include an answer-logit contrast with controlled prompt-template variation for a transformer, a region-level class contrast with validated appearance changes for segmentation, and a conditional denoising-score contrast for a diffusion model. Logits, probes, intermediate representations, and low-rank parameter perturbations provide finite-dimensional coordinates. For total operational dimension $d = d_T + d_N$, dense moment storage costs $O(d^2)$, and direct factorization or eigendecomposition costs $O(d^3)$ per slice. Minibatch accumulation and iterative eigensolvers support larger operational subspaces; feature or score extraction remains dependent on the backbone and input size. These examples are candidate extensions for future evaluation.

**Admission of new nuisance specifications.** A proposed admission protocol prespecifies independent checks of nuisance relevance, task preservation, split stability, and nuisance-block conditioning. An identifiability conclusion requires all four checks to pass; a failed prerequisite is recorded as an unvalidated specification. For a finite admitted family, report the minimum certificate with uncertainty accounting for the family. The present experiments evaluate prespecified nuisance hypotheses, with implemented checks and sensitivity analyses in Appendices B.4, C, and D.5. The general admission procedure remains to be evaluated.

### B.2 Ridge Regularization and the Unregularized Target

For $J = \left(\begin{smallmatrix} A & B \\ B^\top & C \end{smallmatrix}\right) \succeq 0$, with $A = J_{TT}$, $B = J_{TN}$, $C = J_{NN} \succ 0$, and $\lambda_{\mathrm{rid}} > 0$, define

$$S_0(J) = A - BC^{-1}B^\top, \qquad S_{\mathrm{rid}}(J) = A - B(C + \lambda_{\mathrm{rid}} I)^{-1}B^\top. \tag{B.3}$$

Both are positive semidefinite. The resolvent identity, whose inverse factors commute, gives

$$\begin{aligned} \Delta_{\mathrm{rid}}(J) &:= S_{\mathrm{rid}}(J) - S_0(J) = \lambda_{\mathrm{rid}} BC^{-1}(C + \lambda_{\mathrm{rid}} I)^{-1}B^\top \succeq 0, \\ \|\Delta_{\mathrm{rid}}(J)\|_{\mathrm{op}} &\le \frac{\lambda_{\mathrm{rid}} M^2}{\kappa(\kappa + \lambda_{\mathrm{rid}})} \quad \text{if } C \succeq \kappa I,\ \kappa > 0,\ \|B\|_{\mathrm{op}} \le M. \end{aligned} \tag{B.4}$$

Conditioning is essential: scalar blocks $A = 1$, $B = \sqrt{c}$, $C = c > 0$ give a positive-semidefinite joint matrix with $S_0 = 0$ and $S_{\mathrm{rid}} = \lambda_{\mathrm{rid}}/(c + \lambda_{\mathrm{rid}}) \to 1$ as $c/\lambda_{\mathrm{rid}} \downarrow 0$. Comparisons therefore retain the declared standardized nuisance coordinates and ridge value.

### B.3 Error Allowances and Lower Certificates

**Proposition B.1** (Simultaneous lower certificate)**.** *Fix $G$ groups and weights $\pi_g > 0$ summing to one. Within each group, let $M_{gi}$ be i.i.d. with mean $J_g$, $0 \preceq M_{gi} \preceq R_g I_d$, for specified finite $R_g$, and $\widehat{J}_g = n_g^{-1} \sum_i M_{gi}$. For $0 < \delta < 1$, set*

$$\begin{aligned} u_\delta &= \log(2dG/\delta), \qquad t_g = R_g \left( \sqrt{2u_\delta/n_g} + 2u_\delta/(3n_g) \right), \\ \mathcal{E}_\delta &= \bigcap_g \{ \|\widehat{J}_g - J_g\|_{\mathrm{op}} \le t_g \}, \qquad \Pr(\mathcal{E}_\delta) \ge 1 - \delta. \end{aligned} \tag{B.5}$$

*Assume each $C_g \succ 0$. Define observable bounds and allowances by*

$$\begin{aligned} \overline{M}_g &= \|\widehat{B}_g\|_{\mathrm{op}} + t_g, & \underline{\kappa}_g &= \lambda_{\min}(\widehat{C}_g) - t_g, \\ h_{\mathrm{rid}}(t, M) &= t + (2Mt + t^2)/\lambda_{\mathrm{rid}} + M^2 t/\lambda_{\mathrm{rid}}^2, & a_g &= h_{\mathrm{rid}}(t_g, \overline{M}_g), \\ d_g &= \frac{\lambda_{\mathrm{rid}} \overline{M}_g^2}{\underline{\kappa}_g(\underline{\kappa}_g + \lambda_{\mathrm{rid}})} & & \text{when all } \underline{\kappa}_g > 0. \end{aligned} \tag{B.6}$$

*Report the lower certificate in (14) with audit distribution $\pi$ only when these conditioning margins are positive; otherwise record conditioning failure. Its probability of exceeding the unregularized population certificate is at most $\delta$. On $\mathcal{E}_\delta$, a reported level $\underline{I}_{\alpha,\pi} \ge \rho > \varepsilon$ also implies $\sum_{g:\lambda_{\min}(S_0(J_g)) \ge \rho - \varepsilon} \pi_g \ge \alpha$.*

*Proof.* For $X_{gi} = M_{gi} - J_g$, $\|X_{gi}\|_{\mathrm{op}} \le R_g$ and $\mathbb{E}[X_{gi}^2] = \mathbb{E}[M_{gi}^2] - J_g^2 \preceq R_g^2 I$. Matrix Bernstein [26, Theorem 1.4] with the displayed $t_g$ gives failure at most $\delta/G$ per group; the union bound allows dependence between group estimates. On this event, block extraction and Weyl give $\|B_g\|_{\mathrm{op}} \le \overline{M}_g$ and $C_g \succeq \underline{\kappa}_g I$, so (B.4) bounds the ridge gap by $d_g$.

For the estimation error, write $Q = (C + \lambda_{\mathrm{rid}} I)^{-1}$, $\widehat{Q} = (\widehat{C} + \lambda_{\mathrm{rid}} I)^{-1}$, and $\Delta B = \widehat{B} - B$. Positive semidefiniteness and the resolvent identity give $\|Q\|, \|\widehat{Q}\| \le \lambda_{\mathrm{rid}}^{-1}$ and $\|\widehat{Q} - Q\| \le t/\lambda_{\mathrm{rid}}^2$. Decompose the quadratic-block difference as $\Delta B \widehat{Q} \widehat{B}^\top + B\widehat{Q}\Delta B^\top + B(\widehat{Q} - Q)B^\top$. If $\|B\| \le M$, its three norms are bounded by $t(M + t)/\lambda_{\mathrm{rid}}$, $Mt/\lambda_{\mathrm{rid}}$, and $M^2 t/\lambda_{\mathrm{rid}}^2$. Adding the task-block error $t$ proves the bound $h_{\mathrm{rid}}(t, M)$ and hence $a_g$.

Writing $\lambda_g^0 = \lambda_{\min}(S_0(J_g))$, Weyl now yields

$$\lambda_g^0 \geq \lambda_{\min}(S_{\mathrm{rid}}(J_g)) - d_g \geq \lambda_{\min}(S_{\mathrm{rid}}(\widehat{J}_g)) - a_g - d_g. \tag{B.7}$$

Since $S_0(J_g) \succeq 0$, clipping in (14) preserves $\underline{\lambda}_g \leq \lambda_g^0$. Thus monotonicity of CVaR and reciprocation give

$$\underline{I}_{\alpha,\pi} \leq I^0_{\alpha,\pi} := \left[\mathrm{CVaR}_{\alpha,g\sim\pi}((\lambda_g^0 + \varepsilon)^{-1})\right]^{-1}. \tag{B.8}$$

The information-coverage conclusion follows from Appendix A.5. □

Confidence concerns held-out observations; coverage concerns the fixed weights $\pi$. A finite predeclared nuisance family can be included in the union bound over group–specification pairs; the infimum of certificates preserves coverage separately for each specification. Appendix C.4 supplies matrix allowances for frozen soft gates, including random membership normalization, pooling, and support correction, to which the same deterministic argument applies.

### B.4 Implemented Audit and Numerical Comparison

Tables B.1–B.2 audit uncentered feature moments of normalized scalar task and nuisance coordinates $z_i, b_i$, with uniform within-group weights. Separately from the finite-sample proposition, Gaussian/Wald simulation uses $10^5$ moment draws per group at *nominal* 95% confidence. For general normalized weights $w_i$ and $u_i = (z_i^2, z_i b_i, b_i^2)^\top$, draw $m^* = (A^*, B^*, C^*)^\top \sim N(\widehat{m}, \widehat{\Sigma}_{\widehat{m}})$, where

$$\widehat{m} = \sum_i w_i u_i, \qquad \widehat{\Sigma}_{\widehat{m}} = \frac{\sum_i w_i^2 (u_i - \widehat{m})(u_i - \widehat{m})^\top}{1 - \sum_i w_i^2}. \tag{B.9}$$

Uniform weights give sample covariance divided by $n$. Simulation fixes realized coordinate-standardization constants and weights, omitting evaluation-fitted standardization error. Bonferroni tail $\delta_{\mathrm{W}} = 0.05/(3G)$ covers nuisance conditioning, ridge lower information, and an upper ridge gap within a run ($G = 4$ Waterbirds; 18 CivilComments).

Let $J^*$ have blocks $(A^*, B^*, C^*)$, and write $\widehat{F}_g^0 = S_0(\widehat{J}_g)$, $\widehat{F}_g^{\mathrm{rid}} = S_{\mathrm{rid}}(\widehat{J}_g)$. The $\delta_{\mathrm{W}}$-quantiles $C_g^L, L_g^{\mathrm{rid}}, L_g^0$ of $C^*, [S_{\mathrm{rid}}(J^*)]_+, [S_0(J^*)]_+$ are clipped to $[0, \widehat{C}_g]$, $[0, \widehat{F}_g^{\mathrm{rid}}]$, $[0, \widehat{F}_g^0]$, respectively. The $(1 - \delta_{\mathrm{W}})$-quantile $U_g^\Delta$ of the closed-form ridge gap is enlarged to at least its plug-in value. Lower and upper quantiles use lower and higher order statistics, respectively. The implemented lower information is

$$\underline{\lambda}_g^{\mathrm{W}} = \min\{[L_g^{\mathrm{rid}} - U_g^\Delta]_+, L_g^0, \widehat{F}_g^0\}. \tag{B.10}$$

A denominator $C^*$ or $C^* + \lambda_{\mathrm{rid}}$ at or below $10^{-12}$ gives zero for its corresponding Schur value and an infinite gap; otherwise (B.4) is used even for non-PSD draws. An infinite upper gap or $C_g^L \leq 10^{-12}$ forces zero and an uncertifiable status; support, empirical-PSD, and identity checks can also prevent certification. With $\lambda_{\mathrm{rid}} = 10^{-3}$, certificates with superscripts 0, rid, and W aggregate $\widehat{F}_g^0$, $\widehat{F}_g^{\mathrm{rid}}$, and $\underline{\lambda}_g^{\mathrm{W}}$, respectively, using uniform environment weights, $\alpha = 0.95$, $\varepsilon = 10^{-8}$, and fractional CVaR threshold-atom mass.

Table B.1: Same-cohort summaries of 75 default-ridge held-out audits. Conditioning/gap parentheses are within-run nominal Wald lower/upper values; entries are extrema across runs, not joint confidence statements across seeds or methods. Inflation is ridge minus exact plug-in certificate.

| | | Conditioning and ridge gap | | Certificates | | |
|---|---|---|---|---|---|---|
| Scope | Audits | Minimum $J_{NN}$ | Maximum group gap | Maximum inflation | Minimum exact | Minimum Wald lower |
| CivilComments diagnostics | 5 | 1.456 (1.417) | 0.00116 (0.00133) | 0.00018 | 0.4970 | 0.4618 |
| Waterbirds diagnostics | 5 | 0.211 (0.178) | 0.00125 (0.00166) | 0.00071 | 0.1639 | 0.1504 |
| Frozen-feature training | 30 | 0.227 (0.192) | 0.00144 (0.00199) | 0.00062 | 0.1972 | 0.1814 |
| End-to-end training | 35 | 0.349 (0.349) | 0.00252 (0.00300) | 0.00140 | 0.2185 | 0.2004 |
| All audits | 75 | 0.211 (0.178) | 0.00252 (0.00300) | 0.00140 | 0.1639 | 0.1504 |

Every run passed conditioning and satisfied $\underline{I}^{\mathrm{W}}_{0.95} \le \widehat{I}^{0}_{0.95} \le \widehat{I}^{\mathrm{rid}}_{0.95}$; Waterbirds diagnostic values are $0.1504$, $0.1639$, and $0.1646$. Across the five CivilComments diagnostic seeds, the mean ridge and unregularized certificates are $0.4993$ and $0.4991$, respectively. The inequality $\underline{I}^{\mathrm{W}}_{0.95} \le \widehat{I}^{0}_{0.95}$ is enforced by the caps and does not establish confidence coverage. Training audits use fresh models: original training runs lacked the retained environment blocks needed for checkpoint-specific auditing.

For the separate audits in Table B.2, frozen Group DRO + FG maximizes validation certificate within two percentage points of the best validation worst-group accuracy among $\{0.1, 0.3\}$; seeds 17–21 select $(0.3, 0.1, 0.3, 0.3, 0.3)$. Repeated frozen-feature training/audit runs at ridges $10^{-4}, 10^{-3}, 10^{-2}$ give maximum group gaps $0.0001443$, $0.0014374$, $0.0138286$ and Wald upper values $0.0001996$, $0.0019880$, $0.0191165$.

Table B.2: Separate fresh training audits: five-seed means at the stated coefficients. Table B.1 instead includes frozen models selected over the full grid $\{0.01, 0.03, 0.1, 0.3\}$.

| Training setting | Ridge plug-in | Exact plug-in | Wald lower |
|---|---|---|---|
| Frozen FG, $\lambda_{\mathrm{FG}} = 0.03$ | 0.4002 | 0.3999 | 0.3663 |
| Frozen Group DRO + FG, selected from $\{0.1, 0.3\}$ | 0.6254 | 0.6252 | 0.5664 |
| End-to-end Group DRO + FG, $\lambda_{\mathrm{FG}} = 10^{-4}$ | 0.5447 | 0.5442 | 0.4890 |

# C Hidden-Slice Discovery

Appendix D.4 reports this protocol's outcomes. Frozen-gate evaluation recomputes membership normalization and support correction. Prompts, model revision and execution details are in the artifact.

## C.1 Gate Family and Support Constraints

For the frozen standardized 576-dimensional representation $\psi(x)$ and task class $c$, define a logistic gate with parameters $\phi = (a_c, b_c)$:

$$h_{\phi,c}(x) = \sigma(a_c^\top \psi(x) + b_c), \qquad q_i^0 = \frac{h_{\phi,c}(x_i)}{\sum_{j \in \mathcal{D}_c} h_{\phi,c}(x_j)}, \quad \sigma(t) = (1 + e^{-t})^{-1},$$

$$m_{\mathrm{nom}} = \tau_{\mathrm{slice}} n_c, \qquad m_{\mathrm{ESS}} = \min\{n_c, \max\{1, 0.8 m_{\mathrm{nom}}\}\}, \qquad b_{\max} = \min\{1, 2/m_{\mathrm{nom}}\}.$$

Here $\mathcal{D}_c$ contains the current sample's examples with task label $c$, $n_c = |\mathcal{D}_c|$, and $\tau_{\mathrm{slice}} = 0.10$ is within-class mass. Discovery calibrates $b_c$ to this mean membership; evaluation freezes the selected intercept, without re-centering or mass recalibration. The cap denominator is nominal slice size, not the ESS floor.

Support correction uses Euclidean projection of $q^0$ onto the capped simplex only if its maximum exceeds $b_{\max}$. Writing the result as $q^{\mathrm{cap}}$ and $u_i = 1/n_c$, it takes $\widetilde{q} = u + t(q^{\mathrm{cap}} - u)$ with the largest $t \in [0, 1]$ satisfying $\sum_i \widetilde{q}_i^2 \le 1/m_{\mathrm{ESS}}$. Thus $\widetilde{q} \ge 0$, $\sum_i \widetilde{q}_i = 1$, $\max_i \widetilde{q}_i \le b_{\max}$, and $n_{\mathrm{eff}} = (\sum_i \widetilde{q}_i^2)^{-1} \ge m_{\mathrm{ESS}}$. These constraints do not imply nuisance conditioning. Each sample uses its own $n_c$, making $\widetilde{q}$ batch-dependent. Reported mass averages raw selected-class membership; ESS uses $\widetilde{q}$.

**Metrics and controls.** Fisher and high-loss gates report accuracy, uncalibrated binary NLL (temperature one), and post-hoc composition averaged under $\widetilde{q}$, without information pooling. Predicted labels use a zero-logit threshold; calibrated NLL is stored separately. Random controls weight $k = \lceil \tau_{\mathrm{slice}} n_c \rceil$ sampled rows of the Fisher-selected class uniformly; the constant gate uses $1/n_c$. Controls using feature moments and centered covariances reevaluate the predictive-Fisher and high-loss gates without rediscovery.

### C.2 Discovery Objective and Settings

With matrix $M_i$ and same-class pool $\widehat{J}_{\mathrm{pool},c} = n_c^{-1} \sum_{i \in \mathcal{D}_c} M_i$, the ideal constrained objective searches over class–gate pairs:

$$\widehat{J}_{\phi,c}^{(\eta)} = (1-\eta) \sum_i \widetilde{q}_i(\phi) M_i + \eta \widehat{J}_{\mathrm{pool},c},$$
$$(c^\star, \phi^\star) \in \arg\max_{c,\phi} \left\{ \lambda_{\min}\Big( S_{\mathrm{rid}}(\widehat{J}_{\phi,c}^{(\eta)}) \Big) + \varepsilon \right\}^{-1}, \qquad 0 \le \eta < 1. \tag{C.1}$$

Table C.1: Fixed hidden-discovery settings; sensitivity grids are declared before metadata reveal.

| Component | Specification |
|---|---|
| Representation/nuisance | Gate: frozen ImageNet MobileNetV3-small; nuisance: CLIP ViT-B/32 image projection onto a fixed water-minus-land text direction. |
| Head and calibration | Binary linear head: 1,200 full-batch AdamW steps, learning rate $0.01$, weight decay $10^{-4}$; temperature minimizes calibration NLL over 400 log-spaced values in $[0.05, 20]$. |
| Gate optimization | Ten restarts; 1,000 AdamW steps, learning rate $0.03$, weight decay $10^{-4}$; Gaussian initial weights with standard deviation $0.02$. |
| Matrix and stabilization | $M_i = p_i(1-p_i) v_i v_i^\top$ on two task/nuisance coordinates; $\eta = 0.10$, $\lambda_{\mathrm{rid}} = 10^{-3}$, $\varepsilon = 10^{-8}$. |
| Replication/sensitivity | Seeds 17–21; five validation folds; mass and shrinkage grids each $\{0.05, 0.10, 0.20\}$, headline $0.10$. |

For the scalar task/nuisance blocks, write $\widehat{F}^{\mathrm{rid}} = S_{\mathrm{rid}}(\widehat{J}_{\phi,c}^{(\eta)})$. Optimization and checkpoint/restart/class selection use $-\log(\widehat{F}^{\mathrm{rid}} + \varepsilon) - 10(\bar{h}/\tau_{\mathrm{slice}} - 1)^2$, where $\bar{h} = n_c^{-1} \sum_i h_{\phi,c}(x_i)$. The intercept is recalibrated before discovery evaluation; only mass-feasible finite checkpoints are retained, including a constant-gate baseline. High-loss search uses the same gate, mass and support rules, replacing negative log information with weighted, uncalibrated binary NLL. Information uses $v_i = (z_i, b_i)^\top$: $z_i$ is the raw head logit $f_w(x_i)$ standardized by fitting-subset moments, $b_i$ the standardized CLIP nuisance score, and $p_i = \sigma(f_w(x_i)/T)$. Calibration changes the unnormalized predictive factor. Shrinkage pools complete matrices within the current sample's class before taking the Schur complement. Feature controls omit $p_i(1-p_i)$ but retain gate weights. The centered control forms gated and uniform same-class covariances separately, then pools them with weights $1-\eta$ and $\eta$ before taking the Schur complement. For CLIP, normalize five prompt embeddings per concept, normalize their concept means, then normalize the water-minus-land difference. Its dot product with normalized image embeddings gives the raw nuisance score, without background/group annotations.

### C.3 Fitting and Independent Evaluation

Table C.2 uses task labels and official splits throughout; background/group metadata remain sealed until all seeds finish held-out evaluation. Jaccard overlap uses the $k = \max\{1, \lceil \tau_{\mathrm{slice}} n_c \rceil\}$ largest raw memberships in each selected class, intersected on original row identities; different classes give zero. None of the ten Fisher/high-loss gates has a cutoff tie. Fold stability applies this rule to the official training data's 15% calibration subset: average same-class fold-pair overlaps, then weight these class averages by selected-fold counts normalized over classes with at least two gates. Neither comparison uses positive-membership supports or corrected-weight ranks.

### C.4 Held-Out Targets and Uncertainty

Conditional on fitting data and fitted quantities, let $P_c$ be the selected-class distribution and $n = n_c$ its evaluation sample size. For frozen gate $h$ and matrix map $M$,

$$J_h = \frac{A}{p}, \quad A = \mathbb{E}_{P_c}[hM], \quad p = \mathbb{E}_{P_c}[h] > 0; \qquad \widehat{J}_h = \frac{\widehat{A}}{\widehat{p}}, \quad (\widehat{A}, \widehat{p}) = \frac{1}{n} \sum_{i=1}^n (h_i M_i, h_i). \tag{C.2}$$

Table C.2: Fitting and evaluation sequence under the frozen protocol.

| Stage | Fitted or computed | Carried forward |
|---|---|---|
| Training/calibration | Split official training data 85/15 by task label: fit head and coordinate normalizers on 85%; temperature on 15%. CLIP scores use official-training, then fit-subset normalization. | Fixed predictor, coordinates, nuisance direction, temperature. |
| Validation folds | Five shared task-stratified assignments across selectors; discover on complementary folds, evaluate on held-out folds; assess stability on the 15% training-calibration subset. | Protocol assessment; fold gates are not ensembled. |
| Full validation | Refit the gate under the fixed protocol; select class, gate, and restart by discovery utility. | Freeze gate weights, calibrated intercept, selected class, support settings. |
| Test | Apply frozen memberships; recompute admissible weights, same-class pool, and metrics before metadata reveal. | Held-out results, then post-hoc interpretation. |

Support correction leaves all five primary Fisher gates' test weights unchanged; for high-loss seed 19 it raises ESS from $94.48$ to $102.72$. Frozen memberships need not give fixed numerical weights or inactive bootstrap correction.

**Proposition C.1** (Frozen-gate ratio bound)**.** *For i.i.d. evaluation observations from $P_c$, assume $0 \le h \le 1$ and $0 \preceq M \preceq M_{\max} I_d$, with finite $M_{\max}$ specified independently of evaluation data. Given $0 < \delta < 1$, let*

$$u = \log(4d/\delta), \qquad r_A = M_{\max}\left(\sqrt{2u/n} + 2u/(3n)\right), \qquad r_p = \sqrt{\log(4/\delta)/(2n)}.$$

*With conditional probability at least $1-\delta$, simultaneously $\|\widehat{A} - A\|_{\mathrm{op}} \le r_A$ and $|\widehat{p} - p| \le r_p$. Whenever $\underline{p} = \widehat{p} - r_p > 0$ on this event,*

$$\|\widehat{J}_h - J_h\|_{\mathrm{op}} \le r_J := \frac{r_A + \|\widehat{J}_h\|_{\mathrm{op}} r_p}{\widehat{p} - r_p}. \tag{C.3}$$

*Proof.* The centered matrices $h_i M_i - A$ have norm at most $M_{\max}$ and variance bounded by $M_{\max}^2 I_d$. Matrix Bernstein [26, Theorem 1.4] gives the numerator event with failure at most $\delta/2$; Hoeffding gives the denominator event with failure at most $\delta/2$. Their union bound and

$$\widehat{J}_h - J_h = \frac{(\widehat{A} - A) - \widehat{J}_h(\widehat{p} - p)}{p}, \qquad p \ge \widehat{p} - r_p > 0,$$

prove the observable bound. The shared random denominator is therefore explicitly controlled. □

**Pooling and support correction.** Set $J_{\mathrm{pool},c} = \mathbb{E}_{P_c}[M]$ and $J_h^{(\eta)} = (1-\eta)J_h + \eta J_{\mathrm{pool},c}$. Allocate $\delta_{\mathrm{ratio}} + \delta_{\mathrm{pool}} = \delta$, using the former in Proposition C.1. For the evaluation mean $\widehat{J}_{\mathrm{pool},c} = n^{-1}\sum_i M_i$, matrix Bernstein supplies

$$r_{\mathrm{pool}} = M_{\max}\left[\sqrt{\frac{2\log(2d/\delta_{\mathrm{pool}})}{n}} + \frac{2\log(2d/\delta_{\mathrm{pool}})}{3n}\right].$$

Writing $\widetilde{J}_h = \sum_i \widetilde{q}_i M_i$, the triangle inequality gives $\|\widetilde{J}_h - \widehat{J}_h\|_{\mathrm{op}} \le M_{\max}\sum_i |\widetilde{q}_i - q_i^0| =: r_{\mathrm{proj}}$. Hence, on the simultaneous event and when $\underline{p} > 0$,

$$\|(1-\eta)\widetilde{J}_h + \eta\widehat{J}_{\mathrm{pool},c} - J_h^{(\eta)}\|_{\mathrm{op}} \le (1-\eta)(r_J + r_{\mathrm{proj}}) + \eta r_{\mathrm{pool}}. \tag{C.4}$$

The union bound permits dependence between pool and ratio estimates; $r_{\mathrm{proj}}$ covers batch-dependent support correction. If the pool is fixed from fitting data, it belongs to the conditional target and needs no evaluation-error term. Appendix B.3 converts the matrix allowance into a Schur/eigenvalue lower certificate, requiring positive lower membership mass and nuisance-conditioning margins. The target retains shrinkage; evaluation-estimated centering additionally requires first-moment control.

**Reported uncertainty.** Each of 2,000 within-class bootstrap replicates resamples examples with membership, coordinates, logit and label together, recomputing support-corrected weights and the same-class pool. Gate coefficients/intercept, normalizers and temperature stay frozen. The 95% interval uses the 2.5th/97.5th percentiles of the pooled predictive matrix's ridge Schur value, retaining $\eta = 0.10$ and $\lambda_{\mathrm{rid}} = 10^{-3}$, with linear quantile interpolation; $\varepsilon$ enters inverse information separately. For any reported metric $T$, form paired differences $\Delta_s = T_s^{\mathrm{gate}} - T_s^{\mathrm{control}}$ across five seeds, then report $\overline{\Delta} \pm t_{0.975,4} s_\Delta / \sqrt{5}$, with sample standard deviation $s_\Delta$; single-procedure intervals use its five values. Bootstrap intervals condition on fitted quantities; seed intervals reflect shared-benchmark fitting variation. Neither validates the finite-sample radii above.

# D Experimental Protocols and Additional Results

## D.1 Experimental Quantities and Aggregation

Table D.1 identifies the operational targets; Appendix B.1 defines the matrix constructions. Real-data audits use nuisance ridge $10^{-3}$ and stabilization $10^{-8}$; synthetic audits use $\varepsilon = 10^{-12}$. Certificates use $\alpha = 0.95$. Primary predefined-group coordinates use training moments; prospective normalization is specified below. Reported evaluation Loss-CVaR averages group-mean NLLs at or above their empirical 95th percentile, weighting groups equally; synthetic experiments use environment-mean NLLs. The Loss-CVaR training baseline averages the largest $\lceil 0.1n \rceil$ individual-example NLLs.

Table D.1: Diagnostic targets and replication units.

| Experiment | Representation and reported quantity | Seeds |
|---|---|---|
| Prospective Waterbirds | Fine-tuned MobileNetV3-small; background-probe surviving fraction | 22–26 |
| Synthetic | Scalar task $z = s + \delta_e r$ and nuisance $s$; exact population moments, inverse-information tail and certificate | 11–15 |
| Hidden Waterbirds slices | Frozen MobileNetV3-small and fixed CLIP direction; shrunken predictive information | 17–21 |
| CivilComments groups | TF-IDF logistic classifier; feature–Schur certificate | 13–17 |
| Waterbirds groups | Frozen MobileNetV3-small and logistic fitting; feature–Schur certificate | 17–21 |

**Uncertainty.** Prospective, synthetic, predefined-group, and frozen-feature training summaries report nominal 95% normal-approximation half-widths $1.96s/\sqrt{5}$, with sample standard deviation $s$. Hidden-slice and group-free comparisons use five-seed Student-$t$ intervals; contrasts are paired within seed. The end-to-end coefficient sweep reports standard deviations, and the conditional Gaussian/Wald audit is separate (Appendix B.4). Hidden-gate bootstrap intervals condition on the frozen gate and recompute evaluation weights (Appendix C). Acquisition retains means without recovered per-seed intervals.

Seeds describe fitting/partition or control variation on fixed benchmark splits, not independent datasets. For the predefined-group Waterbirds diagnostic in Appendix D.5, the classifier is deterministic after feature extraction and logistic fitting; only its random controls and reserve anchor vary across seeds. Prospective intervals use training seeds: 23,176 composites share 5,794 birds, with two donors per bird and arm. Accuracy changes are percentage points; information retains its original units.

## D.2 Prospective Nuisance Intervention

**Training and frozen selection.** ImageNet-initialized MobileNetV3-small [8, 22] is fully fine-tuned on Waterbirds [23] for seeds 22–26: 50 epochs, batch size 32, learning rate $10^{-4}$, and bf16 on an H100. A validation-fitted temperature and class-balanced logistic background probe use the pooled representation. Five probe folds are stratified by validation group (3,000 maximum iterations); features use validation standardization. Validation nuisance scores are out of fold and use validation moments; test scores use the full-validation probe and test moments. Task logits use training moments. Test normalization never selects the slice. The rule is

$$\begin{aligned} \mathcal{G}_{\text{eligible}} &= \{g : \operatorname{Acc}_v(g) \geq \operatorname{Acc}_v, \quad \operatorname{NLL}_v(g) \leq \operatorname{median}_h \operatorname{NLL}_v(h)\}, \\ g^\star &\in \operatorname*{arg\,min}_{g \in \mathcal{G}_{\text{eligible}}} \lambda_{\min}(\widehat{F}_g^{\text{rid}})/\widehat{J}_{TT,g}. \end{aligned} \tag{D.1}$$

Here $v$ denotes validation and eligibility uses uncalibrated NLL. Temperature scaling [7] minimizes full-validation binary NLL over 400 log-spaced temperatures in $[0.05, 20]$, choosing the smaller temperature at ties. The checkpoint, temperature, probe, slice and audit constants are frozen before scoring; checkpoint mismatches abort evaluation. Table D.2 shows that LL (landbird/land) and WW (waterbird/water) qualify; WW is selected in all five seeds.

**Foreground-preserving construction.** Waterbirds combines CUB birds [29] with Places backgrounds [30] under soft CUB mattes. For each bird, we retain background pixels verbatim outside the

Table D.2: Validation selection. Means ± nominal 95% normal-approximation half-widths across seeds 22–26; accuracy is a fraction. Information uses uncentered feature moments.

*(a) Performance and eligibility*

| Group | $n$ | Accuracy | NLL | Eligible |
|---|---|---|---|---|
| Landbird/land | 467 | $0.990 \pm 0.005$ | $0.038 \pm 0.016$ | Yes |
| Landbird/water | 466 | $0.703 \pm 0.019$ | $1.670 \pm 0.156$ | No |
| Waterbird/land | 133 | $0.390 \pm 0.021$ | $7.718 \pm 0.800$ | No |
| Waterbird/water | 133 | $0.920 \pm 0.013$ | $0.638 \pm 0.038$ | Yes |

*(b) Information*

| Group | Raw information | Projected information | Surviving fraction |
|---|---|---|---|
| Landbird/land | $0.737 \pm 0.046$ | $0.436 \pm 0.045$ | $0.590 \pm 0.027$ |
| Landbird/water | $0.794 \pm 0.032$ | $0.289 \pm 0.014$ | $0.365 \pm 0.026$ |
| Waterbird/land | $0.973 \pm 0.033$ | $0.764 \pm 0.031$ | $0.786 \pm 0.036$ |
| Waterbird/water | $2.063 \pm 0.054$ | $0.511 \pm 0.021$ | $0.248 \pm 0.015$ |

dilated foreground and fill the covered region using Telea inpainting [25], with a 5-pixel inpainting radius and a $5 \times 5$ elliptical dilation kernel. Its extracted bird layer is recomposed onto training-image donors of the same or opposite land/water background type. Donors have foreground fraction at most $0.08$; a deterministic sequence tries four, taking the first with visible inpainted area at most $0.03$, or the least-inpainted of four. The same-type arm undergoes identical recomposition.

Both arms use the same foreground layer, matte, label and source birds. Opaque foreground pixels are unchanged before JPEG encoding (quality 95, no chroma subsampling); soft-edge mixtures vary with background. Two draws per arm yield 23,176 composites. Visible inpainted area has mean $4.13\%$, p95 $8.45\%$, maximum $20.37\%$, and arm means $4.13\%$/$4.12\%$. Correctness is averaged across donors within bird, then across birds.

**Outcomes and confident flips.** Table D.3 reports all groups. The confident half uses the within-slice median calibrated true-label probability among originally correct birds, including ties. In WW, opposite-type/control flip rates are $56.4\%$/$3.3\%$ overall and $41.8\%$/$0.7\%$ in that half; the fixed threshold $p \geq 0.9$ selects no WW birds.

Table D.3: Held-out interventions. Accuracies and flip rates are seed means in percent; controlled drops are same-type minus opposite-type percentage points with nominal 95% normal-approximation half-widths. Panel (b) counts are mean confident-half counts.

*(a) Paired intervention arms*

| Group | Birds | Original | Same type | Opposite type | Controlled drop |
|---|---|---|---|---|---|
| Landbird/land | 2,255 | 98.9 | 99.3 | 72.6 | $26.7 \pm 1.8$ |
| Landbird/water | 2,255 | 71.3 | 72.3 | 99.2 | $-26.9 \pm 2.0$ |
| Waterbird/land | 642 | 42.3 | 40.6 | 92.3 | $-51.7 \pm 2.0$ |
| **Waterbird/water** | 642 | **92.0** | **93.2** | **40.6** | $\mathbf{52.6 \pm 1.4}$ |

*(b) Opposite-type flips among originally correct predictions*

| Group | All originally correct | Confident half | Confident-half count |
|---|---|---|---|
| Landbird/land | 26.8 | 14.8 | 1,122 |
| Landbird/water | 0.2 | 0.1 | 805 |
| Waterbird/land | 1.2 | 0.3 | 136 |
| Waterbird/water | 56.4 | 41.8 | 299 |

**Matched selection comparisons.** Reapplying eligibility within each seed leaves LL and WW. Minimum validation accuracy and maximum uncalibrated NLL, evaluated separately on this identical set, agree with the surviving-fraction selector in all five seeds. All-group comparisons answer a different question; landbird/water determines the model-level certificate (Table D.4).

**Centering and predictive Fisher.** The post hoc check fixes the checkpoints, temperatures, validation coordinates $v = (z, b)^\top$, original scales, eligible sets and intervention outcomes. No model/probe is refitted. With $\langle\cdot\rangle_g$ the divisor-$n_g$ slice mean, compare

$$J_g = \langle vv^\top\rangle_g = \Sigma_g + \mu_g\mu_g^\top, \quad \Sigma_g = \langle (v-\mu_g)(v-\mu_g)^\top\rangle_g, \quad \mu_g = \langle v\rangle_g.$$

Centering does not rescale. The local model $\operatorname{logit} p_{\theta,\xi}(x) = f_w(x)/\widehat{T} + \theta z(x) + \xi b(x)$ at zero has score $(Y-p)v$ with conditional mean zero even when $\mu_g \neq 0$. Its predictive Fisher and the common fraction are

$$I_g = \langle p(1-p)vv^\top\rangle_g, \qquad r_\rho(A) = 1 - \frac{A_{zb}^2}{A_{zz}(A_{bb}+\rho)}, \quad \rho = 10^{-3}.$$

Perturbations follow calibration and profile only the declared background coordinate. The normalized-weight comparator is $K_g = I_g/\bar{q}_g$, $\bar{q}_g = \langle p(1-p)\rangle_g$, so $r_\rho(K_g) = r_{\bar{q}_g\rho}(I_g)$: normalization preserves the unregularized fraction but changes effective ridge. Table D.4 therefore reports it separately.

Table D.4: Selection and matrix sensitivity, evaluated within each seed. Panel (a) distinguishes original eligible and all-group sets; drops are mean percentage points. Panel (b) fixes the original eligible LL/WW set and interventions; fractions/drops are means ± nominal 95% normal-approximation half-widths.

*(a) Matched selectors*

| Statistic | Candidates | Selected group, seeds | Drop |
|---|---|---|---|
| Minimum raw task information | Eligible | Landbird/land, 5/5 | 26.7 |
| Minimum absolute projected information | Eligible | Landbird/land, 5/5 | 26.7 |
| Minimum surviving fraction | Eligible | Waterbird/water, 5/5 | 52.6 |
| Minimum validation accuracy or maximum validation NLL | Eligible | Waterbird/water, 5/5 | 52.6 |
| Minimum normalized-weight fraction | Eligible | Waterbird/water, 5/5 | 52.6 |
| Maximum NLL or minimum accuracy | All four | Waterbird/land, 5/5 | −51.7 |
| Group determining $I^{\mathrm{FG}}_{0.95}$ | All four | Landbird/water, 5/5 | −26.9 |

*(b) Frozen-coordinate constructions*

| Construction | LL fraction | WW fraction | Selected, seeds | Drop |
|---|---|---|---|---|
| Uncentered $J_g$ | $0.590 \pm 0.027$ | $0.248 \pm 0.015$ | WW, 5/5 | $52.6 \pm 1.4$ |
| Centered $\Sigma_g$ | $0.779 \pm 0.028$ | $0.855 \pm 0.032$ | LL, 5/5 | $26.7 \pm 1.8$ |
| Predictive Fisher $I_g$ | $0.987 \pm 0.014$ | $0.273 \pm 0.016$ | WW, 5/5 | $52.6 \pm 1.4$ |
| Normalized weights $K_g$ | $0.987 \pm 0.014$ | $0.269 \pm 0.016$ | WW, 5/5 | $52.6 \pm 1.4$ |

Centering selects LL in all seeds, whereas predictive Fisher retains WW. Slice means therefore contribute to the original ranking: for WW in seed 22, $\mu_z^2 = 2.094$ versus $\Sigma_{zz} = 0.0636$. Table D.5 preserves the per-seed reversal; full means, covariance entries, weights, matrices and hashes are retained in the artifact.

Table D.5: Per-seed fractions on the original global scales. Each cell is LL / WW; the smaller value determines selection. Centering reverses the ranking in every seed.

| Seed | Uncentered $r_\rho(J_g)$ | Centered $r_\rho(\Sigma_g)$ | Predictive $r_\rho(I_g)$ |
|---|---|---|---|
| 22 | 0.5570 / 0.2291 | 0.7430 / 0.8542 | 0.9817 / 0.2587 |
| 23 | 0.6048 / 0.2494 | 0.8098 / 0.8481 | 0.9920 / 0.2729 |
| 24 | 0.6356 / 0.2755 | 0.8132 / 0.8359 | 0.9620 / 0.3032 |
| 25 | 0.5803 / 0.2392 | 0.7514 / 0.9156 | 0.9998 / 0.2571 |
| 26 | 0.5716 / 0.2460 | 0.7779 / 0.8189 | 0.9998 / 0.2732 |

**Development chronology.** The surviving-fraction rule followed an absolute-information pilot on seeds 17–21 and was frozen for newly trained seeds 22–26 on the same benchmark. Primary-cohort entries use the documented rerun; the original-versus-rerun ledger is retained in the artifact. Appendix E.3 gives the Fisher-Glass training comparison under this same intervention.

**Raw, projected, and tail summaries.** We compare four summaries of retained validation geometry: mean raw task second moment $A$, arithmetic mean projected information $P$, reciprocal mean inverse information $H$, and inverse-CVaR certificate $C$. For the four equally weighted groups, let $F_g = \widehat{F}_g^{\mathrm{rid}}$ and $L_g = (F_g + \varepsilon)^{-1}$. Then

$$A = \tfrac{1}{4}\sum_g \widehat{J}_{TT,g}, \qquad P = \tfrac{1}{4}\sum_g F_g,$$

$$H = \left(\tfrac{1}{4}\sum_g L_g\right)^{-1}, \quad C = \left[\mathrm{CVaR}_{0.95,\pi}(L_g)\right]^{-1}, \qquad \pi_g = \tfrac{1}{4}.$$

Thus $H$ is the harmonic mean of the stabilized projected information values $F_g + \varepsilon$. The four summaries use uncentered feature moments; projection uses nuisance ridge $10^{-3}$, and $H$ and $C$ use reciprocal stabilization $\varepsilon = 10^{-8}$. This post hoc analysis joins validation summaries to held-out interventions in the primary (seeds 22–26) and development (17–21) cohorts. It compares summaries of existing models, not training objectives. "Selected FG" denotes the Fisher-Glass models selected within each cohort, including the accuracy-budget fallbacks listed in the caption; Appendix E.3 details the primary training comparison.

Table D.6: Four validation summaries and their intervention associations. Panels (a)–(b): means $\pm$ nominal 95% normal-approximation half-widths, $1.96s/\sqrt{5}$; changes are paired within seed and drops are signed percentage points. Selected FG includes accuracy-budget fallbacks (primary seeds 22, 24; development seed 20). Panel (c): descriptive Spearman coefficients for ten models arranged in five ERM/FG seed pairs per cohort, with average ranks for tied outcomes.

*(a) Model summaries and paired changes*

| Cohort | Model/change | $A$: raw | $P$: projected | $H$: harmonic | $C$: inverse-CVaR |
|---|---|---|---|---|---|
| Primary | ERM | $1.142 \pm 0.012$ | $0.500 \pm 0.010$ | $0.442 \pm 0.011$ | $0.289 \pm 0.014$ |
| | Selected FG | $0.730 \pm 0.086$ | $0.484 \pm 0.047$ | $0.435 \pm 0.043$ | $0.330 \pm 0.043$ |
| | FG − ERM | $-0.412 \pm 0.079$ | $-0.016 \pm 0.046$ | $-0.007 \pm 0.041$ | $+0.041 \pm 0.046$ |
| Development | ERM | $1.152 \pm 0.005$ | $0.489 \pm 0.011$ | $0.413 \pm 0.014$ | $0.255 \pm 0.018$ |
| | Selected FG | $0.754 \pm 0.118$ | $0.519 \pm 0.065$ | $0.433 \pm 0.057$ | $0.303 \pm 0.082$ |
| | FG − ERM | $-0.398 \pm 0.118$ | $+0.030 \pm 0.063$ | $+0.020 \pm 0.066$ | $+0.048 \pm 0.095$ |

*(b) ERM group-selection rules and controlled drops*

| Rule | Candidates | Selected, each cohort | Primary | Development |
|---|---|---|---|---|
| Minimum raw second moment | All four | LL 4/5; LW 1/5 | $+15.8 \pm 21.3$ | $+16.9 \pm 23.2$ |
| Minimum projected / largest inverse contribution / CVaR-active group | All four | LW 5/5 | $-26.9 \pm 2.0$ | $-29.7 \pm 2.2$ |
| Largest inverse contribution | Eligible | LL 5/5 | $+26.7 \pm 1.8$ | $+29.2 \pm 2.8$ |

*(c) Model-level rank associations with held-out outcomes*

| | Primary | | Development | |
|---|---|---|---|---|
| Summary | $D_{\max}$ | $W_{\mathrm{opp}}$ | $D_{\max}$ | $W_{\mathrm{opp}}$ |
| $A$: mean raw | $-0.10$ | $+0.26$ | $+0.26$ | $-0.13$ |
| $P$: mean projected | $-0.09$ | $+0.21$ | $-0.03$ | $-0.43$ |
| $H$: reciprocal mean inverse | $-0.15$ | $+0.21$ | $-0.12$ | $-0.40$ |
| $C$: inverse-CVaR | $+0.15$ | $-0.24$ | $-0.03$ | $-0.48$ |

For Table D.6(c), $D_{\max} = 100 \max_g(\mathrm{Acc}_{\mathrm{same},g} - \mathrm{Acc}_{\mathrm{opposite},g})$ and $W_{\mathrm{opp}} = \min_g \mathrm{Acc}_{\mathrm{opposite},g}$, both over all four groups. Integer correct counts recover exact outcome ties: primary ERM seeds 22 and 25 share drop $348/642$, despite different floating-point subtractions. LL and LW denote landbird/land and landbird/water. Negative drops mean opposite-type recomposition improves accuracy. The group with the largest inverse-information contribution accounts for $38.3\%$ and $40.6\%$ of summed inverse information on average in the primary and development cohorts, respectively. The original headline rules differed: surviving fraction in the primary cohort, absolute projected information in development; this comparison consistently uses all four groups for model summaries and correlations.

Projection and aggregation give different model summaries, with rank associations varying across cohorts. For four equally weighted groups and $\alpha = 0.95$, $C = \min_g F_g + \varepsilon$; its active group also minimizes projected information and maximizes inverse contribution. Here that group is LW, whose opposite-type recomposition improves accuracy. The prospective WW diagnosis instead uses the surviving fraction within the eligible set. This comparison does not establish superior swap-damage prediction by inverse-CVaR; paired changes in $P, H, C$ all have intervals containing zero. The artifact provides per-model and per-seed values, exact-count outcomes, input hashes, and replay code.

**Auxiliary flip prediction.** Each originally correct source bird contributes two opposite-type donor rows. All predictor covariates are evaluated on the original, pre-intervention image. Baseline predictors are uncalibrated NLL, calibrated true-label confidence, absolute raw logit and standardized task logit $z$. Augmentation adds $\beta_g b$ and $|z - \beta_g b|$, with $\beta_g = \widehat{J}_{TN,g}/(\widehat{J}_{NN,g} + 10^{-3})$ frozen from the validation block. Both models use identical five-fold, nonshuffled CUB-species `GroupKFold` splits, keeping species and donor pairs together. Logistic regression uses an intercept, $\ell_2$ penalty, $C = 1$, L-BFGS, tolerance $10^{-4}$, maximum 5,000 iterations, seed 0 and no class weighting or tuning. All columns, including group indicators, are centered and scaled using training-fold means and population standard deviations; standard deviations below $10^{-9}$ are replaced by one.

Pool out-of-fold predictions within seed before computing average precision (reported as AUPRC), area under the receiver operating characteristic curve (AUROC), and mean log likelihood. For seed $s$, $N_s$ counts all retained donor rows in the analyzed population, including flips and nonflips. Let $y_{si}^{\text{flip}} = 1$ indicate an incorrect prediction on the donor composite and $y_{si}^{\text{flip}} = 0$ a correct prediction. The probabilities $q_{si}^A$ and $q_{si}^B$ are the augmented and baseline out-of-fold flip predictions. For clipped probabilities $q \in [10^{-12}, 1 - 10^{-12}]$,

$$G_s = \frac{1}{N_s} \sum_{i=1}^{N_s} \left[ \mathrm{LL}(y_{si}^{\text{flip}}, q_{si}^A) - \mathrm{LL}(y_{si}^{\text{flip}}, q_{si}^B) \right], \quad \mathrm{LL}(y, q) = y \log q + (1 - y) \log(1 - q). \tag{D.2}$$

$G_s$ is augmented-minus-baseline gain in nats per donor row. Table D.7 averages paired changes equally across seeds 22–26. The coordinates improve prediction without group indicators and within WW; with indicators, AUPRC decreases. This tests their incremental value, not unique attribution to the scalar information statistic. Original inputs, exact-source refits and paired records are in the artifact.

Table D.7: Incremental flip prediction. Counts and rates are seed means; parentheses count seeds with improvement. Changes are paired within seed.

| Analysis | Rows | Flip rate | Log-likelihood gain | ΔAUPRC | ΔAUROC |
|---|---|---|---|---|---|
| All groups, no indicators | 9,398 | 0.199 | +0.016 | +0.035 (5/5) | +0.096 (5/5) |
| All groups, with indicators | 9,398 | 0.199 | +0.001 | −0.007 (0/5) | +0.003 (5/5) |
| Selected slice | 1,181 | 0.564 | +0.013 | +0.026 (5/5) | +0.015 (5/5) |

### D.3 Synthetic Classification and Tail Checks

Draw $U_e \sim \mathrm{Uniform}(0, 1)$, set $\delta_e = 0.35 U_e^{1/\gamma}$, and draw independent standard normals $s, r$ for each observation. The task is $z = s + \delta_e r$, $Y = \mathbf{1}\{z > 0\}$, scored by nuisance-only logit $4s$. Thus

$$\Pr(\delta_e < t) = (t/0.35)^\gamma \ (0 \le t \le 0.35), \qquad J_{TT,e} = 1 + \delta_e^2, \quad F_e^0 = \delta_e^2. \tag{D.3}$$

Population feature moments are exact; loss/accuracy use 256 observations per environment. The $\gamma = 1$ experiment uses 300,000 environments and seeds 11–15. Fit the log-survival slope over the largest 5% of inverse information; empirical CVaR averages values at or above the empirical $\alpha$-quantile. Figure A.1 regenerates seed 11 with $L_e = (\delta_e^2 + 10^{-12})^{-1}$, rather than recovering missing original raw output; its survival curve and the five-seed summaries have different replication levels.

Table D.8 checks Theorem 2 and the scalar portfolio law in Appendix A.2. For $K$ independent draws, information $K^{-1} \sum_j \delta_{e,j}^2$ has exponent $K/2$ at $\gamma = 1$; weights affect its constant. The mass sweep

Table D.8: Synthetic checks. Panel (a) and the branch rows are five-seed means ± nominal 95% normal-approximation half-widths; zero widths reflect rounding. Mass-sweep rows are reported single-seed values.

*(a) Prediction and information*

| Quantity | Estimate |
|---|---|
| Mean accuracy | $94.54 \pm 0.00\%$ |
| Fifth-percentile environment accuracy | $88.67 \pm 0.00\%$ |
| $\mathrm{CVaR}_{0.95}$ of environment-mean NLL | $0.262 \pm 0.000$ |
| $I^{\mathrm{FG}}_{0.95}$ | $(1.22 \pm 0.53) \times 10^{-8}$ |
| Inverse-information exponent $\widehat{\beta}$ | $0.498 \pm 0.003$ |

*(b) Inverse-information tail exponents*

| Construction | Setting | Predicted $\beta$ | Estimated $\beta$ |
|---|---|---|---|
| Mass near the null | $\gamma = 1$ | 0.5 | 0.509 |
| Mass near the null | $\gamma = 2$ | 1 | 0.985 |
| Mass near the null | $\gamma = 4$ | 2 | 2.001 |
| Independent branches | $K = 1$ | 0.5 | $0.499 \pm 0.003$ |
| Independent branches | $K = 2$ | 1 | $1.000 \pm 0.009$ |
| Independent branches | $K = 3$ | 1.5 | $1.508 \pm 0.015$ |

predicts $\gamma/2$. An anchor with $\delta_0 = 0.5$ and reserve mass $0.05$ yields information $0.95\delta_e^2 + 0.05(0.25)$, floor $0.0125$, and measured certificate $0.01260 \pm 0.000001$ (five seeds; Appendix E.1).

### D.4 Hidden-Slice Discovery Results

Appendix C gives the blind protocol: task labels, frozen MobileNet features and a fixed CLIP direction are available; background/group metadata remain sealed through held-out evaluation. Results use the primary hidden-slice cohort, seeds 17–21, with full-validation gates frozen before test. Support correction may depend on the test sample but leaves all five primary CLIP gates' test weights unchanged. Information uses same-class pooling; accuracy, uncalibrated NLL and composition use admissible within-class weights without pooling. The artifact identifies this cohort as v2 and records original hashes; the older uncorrected cohort is excluded.

All selectors share classifiers with average accuracy $76.3 \pm 0.7\%$ and post hoc worst-group accuracy $57.3 \pm 1.3\%$. CLIP search finds $98.8\%$ membership-weighted slice accuracy. Table D.9 includes every matched-mass comparator and paired contrast. The random-direction interval and the uncentered and centered feature-moment intervals against high-loss selection contain zero. Feature controls reevaluate gates selected using predictive Fisher without rediscovery.

Table D.9: Hidden-slice comparisons. Panel (a): means $\pm$ 95% Student-$t$ half-widths across five paired seeds; predictive information retains calibration and shrinkage $\eta = 0.10$. Panel (b): CLIP-search minus comparator, with paired 95% Student-$t$ intervals; negative differences indicate lower information.

*(a) Held-out selectors*

| Selector | Predictive projected information | Uncalibrated slice NLL |
|---|---|---|
| Fisher search, fixed CLIP direction | $0.005 \pm 0.001$ | $0.031 \pm 0.017$ |
| High loss, same target within-class mass | $0.019 \pm 0.006$ | $5.779 \pm 0.496$ |
| Random, same target within-class mass | $0.013 \pm 0.001$ | $1.025 \pm 0.059$ |
| Constant class-conditional gate | $0.013 \pm 0.001$ | $1.027 \pm 0.049$ |
| Fisher search, random direction | $0.006 \pm 0.001$ | $0.031 \pm 0.015$ |

*(b) Paired information contrasts*

| Information | Comparison | Difference | 95% interval |
|---|---|---|---|
| Predictive | High-loss slice | $-0.014$ | $[-0.020, -0.008]$ |
| Predictive | Random slice | $-0.008$ | $[-0.009, -0.007]$ |
| Predictive | Constant gate | $-0.008$ | $[-0.009, -0.007]$ |
| Predictive | Random direction | $-0.0004$ | $[-0.0015, 0.0007]$ |
| Feature moments | High-loss slice | $-0.034$ | $[-0.106, 0.039]$ |
| Centered covariance | High-loss slice | $-0.071$ | $[-0.166, 0.024]$ |

**Class selection and overlap.** CLIP, random-direction, random-slice and constant-gate selectors choose landbirds in all five seeds; high loss chooses landbirds in seeds 17,18,21 and waterbirds in 19,20. The two different-class Jaccards are zero; same-class values are $0.00222$, $0.00111$, and $0$. Their mean $0.000666$, reported as $0.001$, combines between-class and within-class differences under Appendix C.3's top-count rule.

**Support and sensitivity.** On same-class rows from the 15% training-calibration subset, cross-fold gate-score correlation is $0.419\,[0.353, 0.485]$ and overlap $0.258\,[0.225, 0.291]$. Test ESS is $414\,[383, 445]$ using admissible weights. Raw membership mass is $0.083\,[0.077, 0.089]$ within the selected class (target $0.10$), not the full test set; landbirds contain 4,510 test examples. Validation cross-fitting, separate from final testing, gives information $0.004$–$0.005$ over masses $\{0.05, 0.10, 0.20\}$ versus high-loss $0.012$–$0.027$, and $0.004$–$0.006$ over shrinkages $\{0.05, 0.10, 0.20\}$.

Composition after held-out evaluation is $97.0\%$ landbird/land for CLIP, $94.4\%$ for random direction, and $58.4\%$ landbird/water for high loss. The hidden slice has lower projected predictive information than the weakest predefined group in all five seeds: mean paired difference $-0.0029$, with 95% Student-$t$ interval $[-0.0034, -0.0024]$, taking the predefined-group minimum separately within each seed. Both sides use the same model, CLIP coordinate, calibrated predictive moments, pooling $\eta = 0.10$, and ridge $10^{-3}$. A metadata-trained oracle probe used only after blind freeze gives

information $0.003 \pm 0.000$, NLL $0.009 \pm 0.004$, and $99.1\%$ landbird/land composition (zero width is rounding). Repair outcomes are in Appendix E.3.

### D.5 Predefined-Group Diagnostics

**Fitting.** CivilComments-WILDS [5, 9] predicts comment toxicity, with $y = 0$ denoting nontoxic comments. Identity-any indicates that at least one of the 24 original identity annotation scores is at least $0.5$. Training-fitted unigram/bigram TF-IDF (50,000 features, minimum document frequency 5, sublinear TF) feeds class-balanced SGD logistic task/identity-any probes ($\ell_2 = 10^{-5}$, 50 passes, tolerance $10^{-3}$). Identity-by-label groups require 100 examples; tail regression uses the largest 50% of group inverse information, requiring eight groups and four tail points. Seed 13 gives the fixed diagnostic; seeds 13–17 give aggregates.

Waterbirds uses frozen ImageNet MobileNetV3-small features (extraction batch 64) and balanced L-BFGS logistic task/background models (1,000 maximum iterations). Four groups are reported directly, without fitting a tail exponent; only controls/anchors vary over seeds 17–21. Primary task/probe and metadata coordinates use training moments. These original runs used an i9-11900F/32 GB workstation, about one minute for CivilComments and three minutes for first Waterbirds extraction/fitting, then cached features.

Table D.10: Original predefined-group diagnostics, distinct from fresh audits in Appendix B.4. Means $\pm$ nominal 95% normal-approximation half-widths; Waterbirds zeros reflect fixed-model fitting.

| Dataset | Average accuracy (%) | Worst-group accuracy (%) | Loss-$\mathrm{CVaR}_{0.95}$ | $I^{\mathrm{FG}}_{0.95}$ |
|---|---|---|---|---|
| CivilComments | $86.10 \pm 0.22$ | $43.75 \pm 0.64$ | $0.998 \pm 0.012$ | $0.500 \pm 0.001$ |
| Waterbirds | $77.11 \pm 0.00$ | $56.54 \pm 0.00$ | $1.741 \pm 0.000$ | $0.165 \pm 0.000$ |

For fixed CivilComments, the weakest-information group is `muslim`/$y = 0$ (accuracy $60.1\%$); lowest accuracy is `black`/$y = 0$ ($44.0\%$). Waterbirds' weakest-information group is landbird/land ($97.3\%$ accuracy), whereas waterbird/land has lowest accuracy ($56.5\%$); inverse-information p95 is $5.90$.

**Specification and stress controls.** The prespecified semantic family is $\{b_{\mathrm{meta}}, b_{\mathrm{probe}}\}$: standardized identity-any/background annotation and its training-fitted probe, without cross-fitting. Semantic relevance follows those annotations; no held-out relevance threshold, anchor task-preservation tolerance, fold-stability admission screen, or nuisance-conditioning threshold was applied. The ridge is $10^{-3}$; family minima are $0.499$ and $0.165$. Reserve anchors (identity-present label-balanced CivilComments; four-group-balanced Waterbirds) are construction inputs, not admission samples. Fresh exact/ridge audits and hidden-gate stability are separate analyses.

Table D.11: Nuisance sensitivity of fixed-model diagnostics. Entries are $I^{\mathrm{FG}}_{0.95}$; worst-group accuracy remains $44.0\%$/$56.5\%$ for CivilComments/Waterbirds.

| Specification | CivilComments | Waterbirds | Role |
|---|---|---|---|
| Metadata | 0.587 | 0.266 | Semantic-family member |
| Learned probe | 0.499 | 0.165 | Semantic-family minimum |
| Random | 0.588 | 0.482 | Negative control |
| Under-projected | 0.588 | 0.482 | Under-projection control |
| Over-projected | 0.001 | 0.012 | Task-erasure control |
| Hidden-loss | 0.060 | 0.177 | Loss-coordinate stress control |

Random nuisance is independent standard normal; under-projection sets it to zero. Over-projection uses standardized task score plus Gaussian noise of standard deviation $0.01$ and deliberately erases task information. "Hidden-loss" is observed test loss with original groups fixed. The random, over-projected, and observed-loss coordinates use evaluated-array normalization and are not independently learned nuisance scores. The metadata/random contrast is $0.587$/$0.588$ on CivilComments, versus probe/random $0.165$/$0.482$ on Waterbirds, illustrating specification dependence.

The versioned artifact (see the artifact availability statement) retains execution commands, full-precision matrices and overlaps, run-history ledgers and expanded source tables, with original cohort provenance. Those records supplement the scientific definitions and controls above.

# E Information-Changing Interventions

This appendix gives the reserve proof, training and acquisition comparisons, and the influence derivative behind Section 6. Distinct cohorts and information targets remain separate. Expanded protocols and numerical breakdowns are retained in the versioned artifact's `docs/extended_appendix_details/`, with executable reproduction guides and provenance (see the artifact availability statement).

## E.1 Information Reserves under Observed-Source Allocation

Fix the predictor, coordinates, and normalization. Let $J_e, J_0 \succeq 0$ have positive-definite nuisance blocks. The anchor is an additional observation source with $F_0^0 := S_0(J_0) \succeq \rho_0 I$, $\rho_0 > 0$. For the allocation in Section 6, each observation's source is observed and allocation probabilities are parameter-independent. Sampling the anchor with probability $r \in (0, 1]$ therefore gives

$$\begin{aligned} J_e^{\mathrm{res}} &= (1-r)J_e + rJ_0, \\ S_0(J_e^{\mathrm{res}}) &\succeq (1-r)S_0(J_e) + rS_0(J_0) \succeq r\rho_0 I. \end{aligned} \tag{E.1}$$

The same statement applies to affine allocation of a fixed operational matrix map; it is not an assertion about an unobserved mixture likelihood.

*Proof.* For $J = \left(\begin{smallmatrix} A & B \\ B^\top & C \end{smallmatrix}\right)$, $C \succ 0$, completing the square gives

$$u^\top S_0(J) u = \min_v q_J(u, v), \qquad q_J(u, v) = \begin{pmatrix} u \\ -v \end{pmatrix}^\top J \begin{pmatrix} u \\ -v \end{pmatrix}, \quad v_* = C^{-1}B^\top u. \tag{E.2}$$

Linearity in $J$ implies

$$\begin{aligned} \min_v \{(1-r)q_{J_e}(u, v) + rq_{J_0}(u, v)\} &\geq (1-r)\min_v q_{J_e}(u, v) + r\min_v q_{J_0}(u, v) \\ &\geq r\rho_0\|u\|^2. \end{aligned} \tag{E.3}$$

The last inequality uses $S_0(J_e) \succeq 0$ and the anchor bound; holding for every $u$, it proves (E.1). □

For any environment distribution, $0 < \alpha < 1$ and $\varepsilon \geq 0$, CVaR monotonicity gives

$$L_e^{\mathrm{res},(\varepsilon)} \leq (r\rho_0 + \varepsilon)^{-1}, \qquad I_\alpha^{\mathrm{FG,res}}(\varepsilon) \geq r\rho_0 + \varepsilon \geq r\rho_0. \tag{E.4}$$

Here $r = m/(n+m)$ for $m$ anchor and $n$ base observations. An estimated anchor's positive lower bound from Appendix B.3 yields the same guarantee on its confidence event. The claim concerns the fixed predictor; retraining requires a new audit. Appendix D.3 evaluates $r = .05$, $\rho_0 = .25$, and the resulting floor .0125.

## E.2 Regularized Training

For a declared group matrix $\widehat{J}_g(w)$, put $\widehat{L}_g(w) = [\lambda_{\min}(S_{\mathrm{rid}}(\widehat{J}_g(w))) + \varepsilon]^{-1}$. With fixed $\pi_g \geq 0$, $\sum_g \pi_g = 1$, training solves

$$\min_{w,\tau} \widehat{R}_{\mathrm{base}}(w) + \lambda_{\mathrm{FG}} \left[\tau + \frac{1}{1-\alpha}\sum_g \pi_g(\widehat{L}_g(w) - \tau)_+\right], \tag{E.5}$$

Here $w$ denotes predictor parameters, $\widehat{R}_{\mathrm{base}}$ is the ERM or Group DRO task-risk objective, and $\lambda_{\mathrm{FG}}$ its information-penalty coefficient. Frozen-feature Waterbirds uses uncentered second moments of standardized task logits and fixed background-probe scores, ridge $10^{-3}$ and $\varepsilon = 10^{-8}$. Training $\alpha = .75$ and audit $\alpha = .95$ both give the maximum inverse information over four equally weighted groups.

**Protocol and selection.** ImageNet MobileNetV3-small features and background-probe scores use training-moment standardization. A linear head receives 1,200 full-batch AdamW updates ($\mathrm{lr} = .01$, weight decay $10^{-4}$, seeds 17–21), Gaussian $N(0, .01^2)$ initial weights and zero intercept. Current logits use full-training population standard deviation plus $10^{-6}$; training projected information is floored at $\varepsilon$ before reciprocal stabilization. Evaluation uses fitted training-logit moments. Group DRO minimizes the maximum group mean cross-entropy; Loss-CVaR uses the largest $\lceil .1n \rceil$ training losses, and JTT refits with weight 20 on initial ERM mistakes.

For each FG family and seed, search $\Lambda_{\mathrm{FG}} = \{.01, .03, .1, .3\}$. Among candidates whose validation worst-group accuracy is within .02 of the family's best, maximize the validation certificate, breaking ties by higher validation worst-group accuracy. This uses validation group annotations. Historical choices are .03 for ERM + FG in all seeds and .1 or .3 for Group DRO + FG. Table E.1 retains all six methods; official test data enter only final evaluation.

Table E.1: Original frozen-feature training cohort, seeds 17–21. Both panels give means ± nominal 95% normal-approximation half-widths; accuracy is percent and the uncentered feature–Schur ridge certificate uses $\alpha = .95$.

| Method | Avg. acc. (%) | Worst acc. (%) | Loss-CVaR$_{.95}$ |
|---|---|---|---|
| ERM | $77.2 \pm 0.1$ | $55.7 \pm 0.3$ | $1.969 \pm 0.007$ |
| Group DRO | $80.9 \pm 0.3$ | $67.3 \pm 0.3$ | $1.143 \pm 0.021$ |
| JTT | $77.4 \pm 0.1$ | $54.1 \pm 0.2$ | $2.167 \pm 0.004$ |
| Loss-CVaR | $78.2 \pm 0.1$ | $52.1 \pm 0.3$ | $2.427 \pm 0.041$ |
| ERM + Fisher-Glass | $80.4 \pm 0.2$ | $56.3 \pm 0.3$ | $1.866 \pm 0.025$ |
| Group DRO + FG | $80.6 \pm 0.9$ | $69.4 \pm 1.3$ | $0.855 \pm 0.082$ |
| **Method** | **Ridge certificate** | **Inverse p95** | |
| ERM | $0.198 \pm 0.001$ | $4.86 \pm 0.02$ | |
| Group DRO | $0.233 \pm 0.003$ | $4.10 \pm 0.06$ | |
| JTT | $0.234 \pm 0.001$ | $4.13 \pm 0.01$ | |
| Loss-CVaR | $0.233 \pm 0.002$ | $4.17 \pm 0.03$ | |
| ERM + Fisher-Glass | $0.399 \pm 0.006$ | $2.46 \pm 0.04$ | |
| Group DRO + FG | $0.614 \pm 0.059$ | $1.63 \pm 0.17$ | |

FG doubles ERM's certificate; adding it to Group DRO raises the certificate from .233 to .614 and mean worst-group accuracy from 67.3% to 69.4%. These historical summaries differ from Appendix B's fresh ridge/exact audit cohort, which does not retroactively certify the historical checkpoints. Their original per-seed outputs were not recovered; retained reports, complete settings and independent rerun commands are in the artifact (see the artifact availability statement).

### E.3 Extended Training and Selection Results

**End-to-end Group DRO + Fisher-Glass coefficient sweep.** MobileNetV3-small is fine-tuned for 10 epochs (batch 128, evaluation batch 512, seeds 17–21) with AdamW learning rate and weight decay $10^{-4}$; the final epoch is evaluated. The reported H100 configuration uses bf16 and TF32. The nuisance is training-standardized background annotation; task logits use minibatch population standard deviation plus $10^{-6}$. Groups with fewer than two observations are omitted. Ridge is $10^{-3}$, $\alpha = .95$, and projected information is floored at $10^{-8}$ before another $10^{-8}$ enters the reciprocal. Table E.2 retains the complete coefficient sweep: stronger penalties can increase information while degrading prediction.

Table E.2: End-to-end Waterbirds Group DRO + Fisher-Glass coefficient sweep. FG entries are means ± sample standard deviations over five seeds, not confidence intervals. The Group DRO reference is the baseline from the $\lambda_{\mathrm{FG}} = 10^{-4}$ sweep; each sweep reran its baseline, so this is not a shared paired reference for every coefficient.

| Objective / $\lambda_{\mathrm{FG}}$ | Avg. acc. (%) | Worst acc. (%) | Loss-CVaR$_{.95}$ |
|---|---|---|---|
| Group DRO reference | 85.4 | 53.2 | 0.925 |
| $5 \times 10^{-5}$ | $79.7 \pm 6.2$ | $56.6 \pm 5.8$ | $0.733 \pm 0.061$ |
| $10^{-4}$ | $79.5 \pm 10.2$ | $57.7 \pm 8.3$ | $0.733 \pm 0.110$ |
| $3 \times 10^{-4}$ | $69.9 \pm 9.7$ | $58.8 \pm 8.7$ | $0.653 \pm 0.064$ |
| $10^{-3}$ | $62.7 \pm 7.9$ | $49.9 \pm 1.8$ | $0.710 \pm 0.012$ |
| $3 \times 10^{-3}$ | $57.7 \pm 4.5$ | $48.6 \pm 6.6$ | $0.717 \pm 0.036$ |
| $10^{-2}$ | $47.9 \pm 16.5$ | $22.7 \pm 16.7$ | $1.031 \pm 0.416$ |
| Objective / $\lambda_{\mathrm{FG}}$ | Ridge certificate | Inverse p95 | |
| Group DRO reference | 0.259 | 3.68 | |
| $5 \times 10^{-5}$ | $0.393 \pm 0.059$ | $2.49 \pm 0.34$ | |
| $10^{-4}$ | $0.502 \pm 0.166$ | $2.10 \pm 0.58$ | |
| $3 \times 10^{-4}$ | $0.614 \pm 0.189$ | $1.71 \pm 0.46$ | |
| $10^{-3}$ | $0.786 \pm 0.124$ | $1.29 \pm 0.20$ | |
| $3 \times 10^{-3}$ | $0.817 \pm 0.068$ | $1.22 \pm 0.10$ | |
| $10^{-2}$ | $0.809 \pm 0.076$ | $1.23 \pm 0.12$ | |

At $\lambda_{\mathrm{FG}} = 10^{-4}$ the certificate rises from .259 to .502 and inverse p95 falls from 3.68 to 2.10; at $\lambda_{\mathrm{FG}} = 10^{-2}$ worst-group accuracy is only 22.7%. Historical report summaries support this table; Appendix B's fresh audits are a separate cohort.

**Prospective information training.** Use Appendix D.2's shared model, 50-epoch schedule and seeds 22–26. Backbone fine-tuning uses AdamW with learning rate and weight decay $10^{-4}$. FG adds coefficients $\{.003, .01, .03\}$ and a background probe trained on detached features (AdamW learning rate .001, default weight decay .01), whose scores are detached in the penalty. After one penalty-free epoch, training uses $\alpha = .75$, ridge .001, batch standard-deviation guard $10^{-6}$, groups of at least six observations with nuisance standard deviation at least .05, and a .05 information floor before direct inversion. This floor differs from audit stabilization.

For this prospective 50-epoch cohort, select the final model with largest validation certificate among candidates whose overall validation accuracy is at least ERM minus .01. If none qualify, maximize the certificate over all FG candidates for that seed and record the fallback. Choices are $[.01, .03, .01, .003, .01]$, with fallbacks in seeds 22 and 24. All audit quantities are frozen before intervention scoring. In the prospective replication, FG raises the validation certificate from .289 to .330, but the controlled drop is 52.9 versus 52.6 percentage points for ERM and the originally-correct flip rate is 57.7% versus 56.4%. The accuracy constraint fails in two seeds and background sensitivity persists.

**Constrained hidden-slice repair.** The primary hidden-slice cohort of Appendix D.4 compares loss, Fisher and combined repair without opening group metadata. Each arm adds its repair penalties to ERM, includes the unchanged model, and must preserve overall, balanced and each-class validation accuracy within .02 of the base. Loss coefficients are $\{.1, .3, 1\}$, Fisher coefficients $\{.01, .03, .1\}$, and combined candidates use their Cartesian product. The same warm start and fresh AdamW optimizer (lr = .01, weight decay $10^{-4}$, gradient clip 10) receive at most 1,200 steps. Calibration NLL at temperature one is checked every 25 steps, with patience 12 and improvement threshold $10^{-5}$; the best checkpoint is restored, with the initial model eligible for selection. The matched ERM continuation sets both penalties to zero. Base temperature remains fixed for information training, selection and subsequent audits.

For the frozen training slice, the dimensionless Fisher penalty is

$$P_{\mathrm{FG}}(w) = \frac{\max\{F_{\mathrm{init}}, \varepsilon\}}{F(w) + \varepsilon}, \qquad \varepsilon = 10^{-8}, \tag{E.6}$$

Here $F(w)$ is the frozen slice's projected predictive information and $F_{\text{init}}$ its value at the unchanged model, with shrinkage .1, ridge .001, and negative numerical information clipped to zero. Within the accuracy envelope, loss/Fisher repair minimizes frozen-slice loss/maximizes information. Combined selection uses the loss–information Pareto frontier, maximizing the smaller relative improvement; ties favor mean relative improvement, then the smaller coefficient sum normalized by grid maxima, $\lambda_{\text{loss}} + 10\lambda_{\text{Fisher}}$, then earlier index. Fresh rediscovery uses matched gate/fold seeds, so the reported population can change.

Table E.3: Constrained repair on the primary hidden-slice cohort, seeds 17–21. Accuracy, NLL and information are means ± 95% Student-$t$ half-widths; composition is a mean percentage. Panel two audits each arm's rediscovered CLIP-defined slice. The Group DRO oracle uses group annotations.

| Arm | Avg. acc. (%) | Worst acc. (%) | Unchanged |
|---|---|---|---|
| Matched ERM continuation | $76.3 \pm 0.7$ | $57.3 \pm 1.3$ | — |
| Loss repair | $76.3 \pm 0.6$ | $56.9 \pm 1.3$ | $3/5$ |
| Fisher repair | $76.3 \pm 0.7$ | $57.3 \pm 1.3$ | $5/5$ |
| Combined repair | $76.4 \pm 0.9$ | $56.7 \pm 1.6$ | $4/5$ |
| Oracle Group DRO | $80.0 \pm 0.3$ | $64.8 \pm 0.9$ | — |
| **Arm** | **Slice NLL** | **Projected info.** | **LL mass (%)** |
| Matched ERM continuation | $0.031 \pm 0.017$ | $0.005 \pm 0.001$ | 97.0 |
| Loss repair | $0.023 \pm 0.017$ | $0.005 \pm 0.002$ | 97.8 |
| Fisher repair | $0.031 \pm 0.017$ | $0.005 \pm 0.001$ | 97.0 |
| Combined repair | $0.031 \pm 0.017$ | $0.005 \pm 0.001$ | 97.2 |
| Oracle Group DRO | $0.033 \pm 0.018$ | $0.005 \pm 0.002$ | 95.9 |

Fisher repair chooses the unchanged model in every seed. Paired worst-group changes versus matched continuation are $-.4$, .0 and $-.6$ points for loss, Fisher and combined repair, with intervals containing zero. The group-supervised Group DRO oracle reaches $64.8\%$ worst-group accuracy while rediscovered information remains .005. LL denotes landbird/land.

**Post hoc group-free coefficient selection.** This analysis was designed after earlier group/test outcomes were inspected. Its grid $\{0, .01, .03, .1, .3\}$ uses (E.6), not the known-group objective's scale. Training is partitioned $70/15/15\%$ for fitting/calibration/gate discovery; a frozen train-only slice supplies the penalty. All candidates share the warm start, training seed, optimizer, 1,200-step budget, checkpoint rule and base temperature above. Official validation enters only after heads are fixed; group metadata stays sealed through blind test evaluation.

Let $\widehat{R}_\lambda$ be mean fixed-temperature validation NLL, $s_\lambda = \text{sd}_i(\ell_{\lambda i})/\sqrt{1199}$ its within-run standard error, and $\lambda_R = \arg\min_\lambda(\widehat{R}_\lambda, \lambda)$ lexicographically. Here $\ell_{\lambda i}$ is the fixed-temperature validation NLL of example $i$, and $\text{sd}_i$ is the sample standard deviation with divisor 1198. Five task-stratified validation folds rediscover candidate-specific slices; $V_{k,\lambda}$ is held-out inverse predictive information and $s_{V,\lambda}$ its sample standard deviation. Selection uses

$$\begin{aligned} \Lambda_{\text{risk}} &= \{\lambda : \widehat{R}_\lambda \le \widehat{R}_{\lambda_R} + s_{\lambda_R}\}, \quad U^{\text{adj}}_\lambda = \overline{V}_\lambda + t_{4,.99} s_{V,\lambda}/\sqrt{5}, \\ \widehat{\lambda}_{\text{hidden}} &= \min \arg \max_{\lambda \in \Lambda_{\text{risk}}} (U^{\text{adj}}_\lambda)^{-1}, \quad t_{4,.99} = 3.74695. \end{aligned} \tag{E.7}$$

The risk allowance uses the minimizer's SE for all candidates. Bonferroni adjustment covers five coefficients, but overlapping gate-training folds make this a ranking score without independent-fold coverage. Accuracy screens and paired inverse-information tests are diagnostics, not selection constraints. The group-supervised comparator maximizes validation worst-group accuracy over the same $\Lambda_{\text{risk}}$, breaking ties toward smaller $\lambda$.

Matched zero continuation restores step zero in all five seeds, explaining equality with frozen ERM despite intervening optimization. Group-free selection equals plain NLL minimization in every seed; the information criterion does not change any choice. Versus matched continuation, paired mean changes are $+2.1$ average-accuracy points ($[+.2, +4.0]$), $+.006$ information ($[+.001, +.010]$), and $-12.6$ worst-group points ($[-21.6, -3.7]$). The group-supervised comparator opens validation annotations only after blind decisions and records are fixed and is not an upper bound. The post

hoc coefficient-selection results and primary repair cohort have separate artifact identifiers (see the artifact availability statement), alongside extended settings and full tables.

Table E.4: Post hoc coefficient selection, five paired seeds. Accuracies are test results; information is projected predictive information in test slices defined by validation-rediscovered, frozen gates. Brackets are 95% Student-$t$ intervals. Policies selecting identical models across these seeds share rows; the text explains both coincidences. Coefficient counts are $\lambda$ : number of seeds.

| Policy | Average accuracy (%) | Worst accuracy (%) |
|---|---|---|
| Frozen ERM / matched zero | 75.5 [74.3, 76.6] | 55.7 [53.3, 58.2] |
| Fixed $\lambda = .03$ | 76.6 [76.1, 77.2] | 51.5 [47.1, 55.8] |
| Risk only / group-free | 77.5 [76.5, 78.6] | 43.1 [33.1, 53.1] |
| Group-supervised comparator | 77.7 [76.6, 78.8] | 44.2 [34.2, 54.3] |
| **Policy** | **Coefficient counts** | **Slice information** |
| Frozen ERM / matched zero | — / 0 : 5 | 0.007 [0.005, 0.009] |
| Fixed $\lambda = .03$ | .03 : 5 | 0.007 [0.004, 0.010] |
| Risk only / group-free | 0 : 1, .3 : 4 | 0.012 [0.006, 0.019] |
| Group-supervised comparator | 0 : 1, .1 : 1, .3 : 3 | 0.011 [0.005, 0.018] |

**Operational choice of the penalty.** For group-free selection, we propose prespecifying an information requirement and a validation-loss budget, then choosing the smallest coefficient satisfying both. If no candidate qualifies, report the requirement as unmet. With neither a specified information requirement nor a group-labeled audit set, retain $\lambda_{\mathrm{FG}} = 0$. A worst-group-accuracy objective requires group labels or a justified group proxy. This threshold-based policy requires prospective validation; the reported experiment maximizes information within an NLL eligibility set.

### E.4 Tail-Transverse Influence

Fix the predictor, coordinates, calibration, slice rules and finite audit weights $\pi_g \geq 0$, $\sum_g \pi_g = 1$. An action specifies a positive-semidefinite path $J_g(t, a) = J_g + tH_{g,a} + o(t)$ as $t \downarrow 0$, with symmetric $H_{g,a}$ and an operator-norm remainder. Allocation, normalization and retained pooling belong in this path; audit weights stay fixed.

**One differential for exact and ridge targets.** Write $J_g = \left(\begin{smallmatrix} A_g & B_g \\ B_g^\top & C_g \end{smallmatrix}\right)$. Choose a fixed $\rho \geq 0$ with $C_g + \rho I \succ 0$: $\rho = 0$ gives exact Schur information and $\rho = \lambda_{\mathrm{rid}}$ gives the ridge target. Set $Q_g = (C_g + \rho I)^{-1}$, $S_\rho(J_g) = A_g - B_g Q_g B_g^\top$, and let $\lambda_g$ be its smallest eigenvalue. Since $DQ_g[H_{NN}] = -Q_g H_{NN} Q_g$,

$$\begin{aligned} DS_\rho(J_g)[H] &= H_{TT} - H_{TN} Q_g B_g^\top - B_g Q_g H_{TN}^\top \\ &\quad + B_g Q_g H_{NN} Q_g B_g^\top, \\ p_g(u) &= \begin{pmatrix} u \\ -Q_g B_g^\top u \end{pmatrix}, \qquad u^\top DS_\rho(J_g)[H] u = p_g(u)^\top H p_g(u). \end{aligned} \tag{E.8}$$

For a simple minimum eigenvalue with unit eigenvector $u_g$, differentiating the eigenvalue equation therefore gives

$$h_{g,a} := D_a \lambda_g = p_g^\top H_{g,a} p_g, \qquad p_g = p_g(u_g). \tag{E.9}$$

Let $s_a$ be the candidate's score vector and $M_{g,a}$ its matrix contribution to group $g$. For allocation rate $\eta_g \geq 0$, additive and normalized contributions have different derivatives:

$$\begin{aligned} H_{g,a} = \eta_g s_a s_a^\top &\implies h_{g,a} = \eta_g (p_g^\top s_a)^2, \\ J_g(t,a) = \frac{J_g + t\eta_g M_{g,a}}{1 + t\eta_g} &\implies H_{g,a} = \eta_g (M_{g,a} - J_g), \\ &\qquad h_{g,a} = \eta_g \left[ p_g^\top M_{g,a} p_g - \lambda_g + \rho \|Q_g B_g^\top u_g\|^2 \right]. \end{aligned} \tag{E.10}$$

Indeed $Q_g C_g Q_g = Q_g - \rho Q_g^2$, so $p_g^\top J_g p_g = \lambda_g - \rho \|Q_g B_g^\top u_g\|^2$; the correction vanishes at $\rho = 0$. Per unit allocation, the additive rank-one contribution is the squared nuisance-adjusted residual

$[u_g^\top(s_{T,a} - B_g Q_g s_{N,a})]^2$; at $\rho = 0$, this is the efficient-score residual. Replacement uses the allocated new-minus-removed contribution. For one observation appended to group $g$, its indicator and original sample-count scaling enter $\eta_g$; they cannot be omitted when claiming a normalized-augmentation derivative.

**Tail sensitivity, including nonsmooth cases.** Fix $\varepsilon > 0$ and $0 < \alpha < 1$, and put $L_g = (\lambda_g + \varepsilon)^{-1}$. Appendix A.5's finite CVaR dual uses $\mathcal{W}_{\alpha,\pi} = \{\omega \geq 0 : \sum_g \omega_g = 1,\ \omega_g \leq \pi_g/(1-\alpha)\}$. Let $\Omega^\star_\alpha(L)$ maximize $\sum_g \omega_g L_g$ over this set. If the maximizer $\omega_\alpha$ is unique and eigenvalues are simple,

$$G_\alpha(a) := -D_a \operatorname{CVaR}_{\alpha,\pi}(L) = \sum_g \omega_{\alpha,g} \frac{h_{g,a}}{(\lambda_g + \varepsilon)^2}, \qquad D_a I_{\alpha,\pi} = I^2_{\alpha,\pi} G_\alpha(a). \tag{E.11}$$

Thus $G_\alpha(a)$ is the first-order decrease in inverse-information CVaR along action $a$'s specified allocation path. It follows from $D_a L_g = -h_{g,a}/(\lambda_g + \varepsilon)^2$; the dual weights include fractional threshold mass and upper-tail normalization.

For a repeated minimum eigenvalue with eigenspace $\mathcal{E}_g$, symmetric eigenvalue perturbation gives the first formula below. At a CVaR tie, the right derivative of a maximum is the maximum of its active directional derivatives; negation gives the second:

$$\begin{aligned} h_{g,a} &= \min_{\substack{u \in \mathcal{E}_g \\ \|u\|=1}} p_g(u)^\top H_{g,a} p_g(u), \\ G^+_\alpha(a) := -D^+_a \operatorname{CVaR}_{\alpha,\pi}(L) &= \min_{\omega \in \Omega^\star_\alpha(L)} \sum_g \omega_g \frac{h_{g,a}}{(\lambda_g + \varepsilon)^2}. \end{aligned} \tag{E.12}$$

These formulas cover either or both sources of nondifferentiability; the reciprocal-certificate identity holds with right derivatives. An arbitrary fixed maximizing tail vector is only a score unless it attains this direction-dependent minimum.

For assigned label $y_a$, tail-transverse influence multiplies the directional information gain by current prediction loss:

$$\operatorname{TTI}(a) = G_\alpha(a) \ell_w(x^{\mathrm{cf}}_a, y_a), \qquad \ell_w(x, y) = -\log p_w(y \mid x), \tag{E.13}$$

using $G^+_\alpha$ at nonsmooth points. This ranks actions at the fixed predictor; finite augmentation, refitting and changed normalization require a new audit. The implemented residual-square surrogate $\widetilde{G}_\alpha$ below omits allocation/baseline terms and is distinguished from this derivative.

### E.5 Feature-Space Acquisition Experiments

**Candidates, budgets and refitting.** Using training-standardized frozen MobileNetV3-small representations, let $\bar{v}_{y,b}$ denote the official-training mean for task label $y$ and background $b \in \{0, 1\}$. Each official-validation candidate becomes

$$v^{\mathrm{cf}} = v + \gamma_{\mathrm{cf}}(\bar{v}_{y,1-b} - \bar{v}_{y,b}), \qquad y^{\mathrm{cf}} = y, \quad b^{\mathrm{cf}} = 1 - b. \tag{E.14}$$

This preserves the assigned label, not necessarily semantic or pixel-level content. The reported unit-strength run uses label-conditional means for every candidate. A balanced logistic background probe, fitted and standardized on training data, stays fixed and scores the translated features. Test data supply final evaluation only.

Budgets $b_{\mathrm{acq}} = .01, .02, .05$ use $k = \min\{\max(1, \operatorname{round}(b_{\mathrm{acq}} n_{\mathrm{train}})), n_{\mathrm{cand}}\}$, giving $48, 96, 240$ candidates from the 1,199-example validation pool relative to 4,795 original training examples. These are formula-implied counts; historical selected-index records were not recovered. Acquired features and labels augment training, and a balanced logistic task head is fitted from scratch, without an FG penalty or special acquisition weights. Both probe and head use L-BFGS, at most 1,000 iterations, and library defaults ($C = 1$, tolerance $10^{-4}$, fitted intercept). Class weights are recomputed on the augmented set. New task logits are normalized by augmented-training moments, while nuisance normalization stays fixed. Acquired validation examples are no longer an independent evaluation sample. Seeds are 17–21; the historical scikit-learn version was not pinned.

**Implemented score and its scope.** Let $(z, q)$ be the task logit and nuisance score standardized on original training data. Original validation-group moments are unweighted second moments, without $p(1-p)$; put $\beta_g = J_{TN,g}/(J_{NN,g} + .001)$, $\lambda_g = J_{TT,g} - J_{TN,g}^2/(J_{NN,g} + .001)$, and $L_g = (\lambda_g + \varepsilon)^{-1}$ with $\varepsilon = 10^{-8}$. Numerical negative information is floored at zero. For the threshold-inclusive tail $T = \{g : L_g \geq \text{quantile}_{.95}(L_1, \ldots, L_4)\}$, the code ranks by

$$\widetilde{G}_\alpha(a) = \frac{\sum_{g \in T}(\lambda_g + \varepsilon)^{-2}(z_a^{\text{cf}} - \beta_g q_a^{\text{cf}})^2}{\max\{\sum_{g \in T}(\lambda_g + \varepsilon)^{-2}, 10^{-12}\}}, \qquad \widetilde{\text{TTI}}_\alpha(a) = \widetilde{G}_\alpha(a)\ell_w(v_a^{\text{cf}}, y_a). \tag{E.15}$$

Every tail projector is applied to every candidate, irrespective of its assigned group; for a unique worst group the inverse-square weight cancels. The score lacks group allocation, sample-count scaling, old-information subtraction and replacement terms. Its nonnegative residual square is an unnormalized rank-one contribution at fixed coordinates, not the derivative of normalized augmentation or of the post-refit certificate. Threshold-inclusive ties also differ from the fractional/direction-dependent CVaR convention in Appendix E.4.

High loss uses $\ell_w$; pure tail selects by $\widetilde{G}_\alpha$ alone. Rank product maps ascending `argsort` positions for each score evenly from 0 to 1, then multiplies their rank scores; equal values receive separate ranks. Uncertainty product multiplies $\widetilde{G}_\alpha$ by $p(1-p)$. The acquisition baseline has $78.2\%$ average and $52.5\%$ worst-group accuracy, certificate .209, and weakest-information group landbird/land; it is distinct from the predefined-diagnostic cohort.

Table E.5: Feature-space acquisition, means over seeds 17–21. $\Delta a$ is worst-group accuracy gain in percentage points and $\Delta I$ the absolute ridge-certificate gain. Panel (a) gives the complete strength-by-budget comparison; panel (b) retains pure-tail and alternative-product controls at unit strength. Only aggregate reports survive, so historical seed-level intervals are unavailable.

(a) Strength and budget: implemented geometric score × loss versus high loss

| | | $\widetilde{\text{TTI}}_\alpha$ | | High loss | |
|---|---|---|---|---|---|
| $\gamma_{\text{cf}}$ | Budget | $\Delta a$ | $\Delta I$ | $\Delta a$ | $\Delta I$ |
| 0.25 | 1% | +4.05 | +0.0308 | +5.76 | +0.0411 |
| | 2% | +7.17 | +0.0366 | +7.63 | +0.0477 |
| | 5% | +8.88 | +0.0521 | +10.12 | +0.0590 |
| 0.5 | 1% | +6.39 | +0.0205 | +4.36 | +0.0328 |
| | 2% | +5.61 | +0.0300 | +4.83 | +0.0446 |
| | 5% | +6.54 | +0.0417 | +8.57 | +0.0437 |
| 1 | 1% | +4.52 | +0.0317 | +3.27 | +0.0603 |
| | 2% | +6.85 | +0.0452 | +5.76 | +0.0581 |
| | 5% | +7.94 | +0.0606 | +7.01 | +0.0630 |
| 1.5 | 1% | +7.48 | +0.0662 | +7.63 | +0.1018 |
| | 2% | +8.72 | +0.0826 | +9.81 | +0.1006 |
| | 5% | +9.97 | +0.1159 | +12.31 | +0.1137 |

(b) Unit-strength controls

| | Pure tail | | Rank product | | Score $\times p(1-p)$ | |
|---|---|---|---|---|---|---|
| Budget | $\Delta a$ | $\Delta I$ | $\Delta a$ | $\Delta I$ | $\Delta a$ | $\Delta I$ |
| 1% | +0.62 | +0.0045 | +5.45 | +0.0287 | +1.56 | +0.0012 |
| 2% | +2.80 | +0.0098 | +6.54 | +0.0326 | +1.71 | +0.0073 |
| 5% | +3.58 | +0.0340 | +6.07 | +0.0516 | +3.12 | +0.0230 |

At $\gamma_{\text{cf}} = 1$, the loss product exceeds high loss in mean worst-group accuracy at all budgets, while high loss gives larger certificate gains. This advantage is strength-dependent: high loss wins accuracy at every budget for .25 and 1.5, and at the $5\%$ budget for .5. The rank product also exceeds the chosen loss product at $1\%$ ($5.45$ versus $4.52$ points); the chosen product is not uniformly best. Information gains and predictive gains are distinct outcomes.

Table E.5 reproduces historical reports, not fresh runs. Missing per-seed JSON/CSV, selected indices and pinned versions preclude uncertainty reconstruction. The artifact supplies rerun code, aggregate candidate score/loss summaries, allocation mechanics and protocols. Five seeds on fixed splits do not constitute five independent datasets.

# F Reproducibility and Evidence Provenance

The accompanying ZIP is the `camera-ready-v22` reproducibility artifact for the final manuscript. It includes the experiment implementations, retained numerical inputs and outputs, final manuscript sources and PDF, and this standalone supplementary document. The versioned release is available at

https://github.com/E-zClap/fisher-glass-learning/releases/tag/camera-ready-v22.

Use the files and checksums in this release when reproducing the reported results. This version adopts numeric citations and harmonizes bibliography typography and link colors in the paper and supplement; no additional evaluation is introduced. The paper and supplement share the complete 30-entry bibliography. Scientific Appendices A–E and the retained experimental results are unchanged. This supplement provides a reader guide, an evidence/replay map, and the retained paired failure-prediction intervals; it introduces no new empirical experiment.

**Two reproduction routes.** *Numerical replay* recomputes tables and comparisons from retained outputs, without retraining image models or obtaining benchmark datasets. *Independent replication* reruns the documented experiments from their original data sources. The latter requires the relevant data, dependencies, and compute resources and need not reproduce historical file bytes. These routes are documented separately in `START_HERE.md`, `docs/REPRODUCING_PRIMARY_EXPERIMENTS.md`, and `docs/REPRODUCING_EXPERIMENTS.md`.

Table F.1: Entry points for the retained evidence. Paths are relative to the ZIP's top-level folder; the experiment map gives exact commands and cohort details.

| Evidence | Included location |
|---|---|
| Prospective intervention and development cohort | `artifacts/prospective_v2/`; `artifacts/prospective_v1/` |
| Matched selectors and matrix constructions | `reanalysis/matched_selectors/`; `reanalysis/matrix_constructions/` |
| Four validation summaries | `reanalysis/four_way_swap/` |
| Within-group failure prediction | `reanalysis/flip_prediction/` |
| Ridge-to-exact audit | `artifacts/w2_audit/`; `scripts/reproduce_w2_audit.py` |
| Corrected hidden-slice discovery and repair | `artifacts/hidden_slice_corrected_v2/` |
| Group-free coefficient selection | `artifacts/group_free_v2/`; `experiments/waterbirds_group_free_lambda/` |
| Worked classification derivation | `reanalysis/classification_mechanism/` |
| Historical training and acquisition summaries | `reports/`; `docs/REPRODUCING_EXPERIMENTS.md` |

**Integrity and cohort identity.** `ARTIFACT_MANIFEST.json` records the distributed files and their SHA-256 hashes, along with original archive identifiers and transformations. The corrected hidden-slice v2 cohort supplies the discovery and repair tables; it must not be replaced by the older uncorrected archive. The prospective development cohort and primary five-seed cohort remain separate. The four-way and matrix comparisons are reanalyses of retained records, not additional independently trained objective arms.

**Historical limits and extended records.** Some original synthetic, frozen-training, and acquisition per-seed outputs were not retained. Their aggregate reports and executable replication workflows are included, but missing historical observations and uncertainty intervals are not reconstructed. Expanded execution details are preserved in `docs/extended_appendix_details/`; those records retain historical numbering, while `docs/MANUSCRIPT_MAP.md` gives the current crosswalk. `docs/REBUTTAL_EXPERIMENT_MAP.md` maps the completed rebuttal additions to their sources and replay routes.

**External assets.** The package includes the two retained composite illustrations used in main-paper Figure 1, with image identifiers, donor metadata, and hashes. Full benchmark datasets, other intervention images, feature caches, and pretrained or fitted model weights are obtained or regenerated through the documented workflows. `DATA_LICENSES.md` records upstream access and rights information; inclusion of the figure illustrations does not assign them a new license.

### F.1 Retained Paired Failure-Prediction Uncertainty

Table F.2: Retained paired uncertainty for auxiliary flip prediction. Changes are augmented minus baseline, paired within each of the five training seeds 22–26 and averaged equally across seeds. Nominal 95% normal-approximation intervals are $\bar{d} \pm 1.96 s_d/\sqrt{5}$, where $s_d$ is the sample standard deviation of the five paired changes. Log-likelihood gains are nats per donor row; AUPRC denotes average precision, and AUROC is the area under the receiver operating characteristic curve. The original species-held-out folds and fitting protocol in Table D.7 are preserved. This table formats the retained summary values; it introduces no new analysis.

| Variant | Metric | Paired mean | Nominal 95% interval |
|---|---|---|---|
| All groups, no indicators | Log likelihood | $+0.0164$ | $[+0.0132, +0.0197]$ |
| | AUPRC | $+0.0354$ | $[+0.0282, +0.0426]$ |
| | AUROC | $+0.0964$ | $[+0.0896, +0.1033]$ |
| All groups, with indicators | Log likelihood | $+0.0009$ | $[+0.0002, +0.0017]$ |
| | AUPRC | $-0.0067$ | $[-0.0085, -0.0048]$ |
| | AUROC | $+0.0027$ | $[+0.0012, +0.0042]$ |
| Selected waterbird/water group | Log likelihood | $+0.0128$ | $[+0.0110, +0.0147]$ |
| | AUPRC | $+0.0257$ | $[+0.0211, +0.0302]$ |
| | AUROC | $+0.0150$ | $[+0.0120, +0.0179]$ |

The augmented predictor adds $\beta_g b$ and $|z - \beta_g b|$ jointly to the same source-image loss, calibrated confidence, and logit predictors. These comparisons therefore assess the two added coordinates together. There is no background-only ablation, and the results do not isolate a residual-only or scalar-information contribution. In the pooled variant with group indicators, the AUPRC change is negative. The intervals summarize training-seed variation on a shared benchmark, not uncertainty across independent datasets.

## References


[1] Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization. *arXiv preprint arXiv:1907.02893*, 2019. doi: 10.48550/arXiv.1907.02893.

[2] Jordan T. Ash, Surbhi Goel, Akshay Krishnamurthy, and Sham M. Kakade. Gone fishing: Neural active learning with Fisher embeddings. In *Advances in Neural Information Processing Systems*, volume 34, pages 8927–8939. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper/2021/hash/4afe044911ed2c247005912512ace23b-Abstract.html.

[3] Jean-Yves Audibert and Alexandre B. Tsybakov. Fast learning rates for plug-in classifiers. *The Annals of Statistics*, 35(2):608–633, 2007. doi: 10.1214/009053606000001217.

[4] Peter J. Bickel, Chris A. J. Klaassen, Ya'acov Ritov, and Jon A. Wellner. *Efficient and Adaptive Estimation for Semiparametric Models*. Johns Hopkins University Press, 1993.

[5] Daniel Borkan, Lucas Dixon, Jeffrey Sorensen, Nithum Thain, and Lucy Vasserman. Nuanced metrics for measuring unintended bias with real data for text classification. In *Companion Proceedings of The 2019 World Wide Web Conference*, pages 491–500. ACM, 2019. doi: 10.1145/3308560.3317593.

[6] D. R. Cox and N. Reid. Parameter orthogonality and approximate conditional inference. *Journal of the Royal Statistical Society: Series B (Methodological)*, 49(1):1–18, 1987. doi: 10.1111/j.2517-6161.1987.tb01422.x.

[7] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, 2017. URL https://proceedings.mlr.press/v70/guo17a.html.

[8] Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, Quoc V. Le, and Hartwig Adam. Searching for MobileNetV3. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1314–1324, 2019. doi: 10.1109/ICCV.2019.00140. URL https://openaccess.thecvf.com/content_ICCV_2019/html/Howard_Searching_for_MobileNetV3_ICCV_2019_paper.html.

[9] Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanas Phillips, Irena Gao, Tony Lee, Etienne David, Ian Stavness, Wei Guo, Berton Earnshaw, Imran Haque, Sara M. Beery, Jure Leskovec, Anshul Kundaje, Emma Pierson, Sergey Levine, Chelsea Finn, and Percy Liang. WILDS: A benchmark of in-the-wild distribution shifts. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 5637–5664. PMLR, 2021. URL https://proceedings.mlr.press/v139/koh21a.html.

[10] Lucien Le Cam. *Asymptotic Methods in Statistical Decision Theory*. Springer Series in Statistics. Springer, 1986. doi: 10.1007/978-1-4612-4946-7.

[11] Evan Z. Liu, Behzad Haghgoo, Annie S. Chen, Aditi Raghunathan, Pang Wei Koh, Shiori Sagawa, Percy Liang, and Chelsea Finn. Just train twice: Improving group robustness without training group information. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 6781–6792. PMLR, 2021. URL https://proceedings.mlr.press/v139/liu21f.html.

[12] El Mustapha Mansouri. Autonomous AI bird feeder for backyard biodiversity monitoring, 2025. URL https://arxiv.org/abs/2508.09398.

[13] El Mustapha Mansouri and Keigo Arai. Fisher Glasses: Tail-certified quantum metrology in quenched environments. arXiv preprint arXiv:2607.01085v1, 2026. URL https://arxiv.org/abs/2607.01085v1.

[14] Whitney K. Newey. Semiparametric efficiency bounds. *Journal of Applied Econometrics*, 5(2): 99–135, 1990. doi: 10.1002/jae.3950050202.

[15] Luke Oakden-Rayner, Jared Dunnmon, Gustavo Carneiro, and Christopher Ré. Hidden stratification causes clinically meaningful failures in machine learning for medical imaging. In *Proceedings of the ACM Conference on Health, Inference, and Learning*, pages 151–159. ACM, 2020. doi: 10.1145/3368555.3384468.

[16] Bhargavi Paranjape, Pradeep Dasigi, Vivek Srikumar, Luke Zettlemoyer, and Hannaneh Hajishirzi. AGRO: Adversarial discovery of error-prone groups for robust optimization. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=IrzkT99fDJH.

[17] Friedrich Pukelsheim. *Optimal Design of Experiments*, volume 50 of *Classics in Applied Mathematics*. SIAM, 2006. doi: 10.1137/1.9780898719109.

[18] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR, 2021. URL https://proceedings.mlr.press/v139/radford21a.html.

[19] Alexandre Rame, Corentin Dancette, and Matthieu Cord. Fishr: Invariant gradient variances for out-of-distribution generalization. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 18347–18377. PMLR, 2022. URL https://proceedings.mlr.press/v162/rame22a.html.

[20] R. Tyrrell Rockafellar and Stanislav Uryasev. Optimization of conditional value-at-risk. *The Journal of Risk*, 2(3):21–41, 2000. doi: 10.21314/JOR.2000.038.

[21] R. Tyrrell Rockafellar and Stanislav Uryasev. Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7):1443–1471, 2002. doi: 10.1016/S0378-4266(02)00271-6.

[22] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. ImageNet large scale visual recognition challenge. *International Journal of Computer Vision*, 115:211–252, 2015. doi: 10.1007/s11263-015-0816-y.

[23] Shiori Sagawa, Pang Wei Koh, Tatsunori B. Hashimoto, and Percy Liang. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations*, 2020. URL https://openreview.net/forum?id=ryxGuJrFvS.

[24] Nimit S. Sohoni, Jared A. Dunnmon, Geoffrey Angus, Albert Gu, and Christopher Ré. No subclass left behind: Fine-grained robustness in coarse-grained classification problems. In *Advances in Neural Information Processing Systems*, volume 33, pages 19339–19352. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper/2020/hash/e0688d13958a19e087e123148555e4b4-Abstract.html.

[25] Alexandru Telea. An image inpainting technique based on the fast marching method. *Journal of Graphics Tools*, 9(1):25–36, 2004. doi: 10.1080/10867651.2004.10487596. URL https://doi.org/10.1080/10867651.2004.10487596.

[26] Joel A. Tropp. User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics*, 12(4):389–434, 2012. doi: 10.1007/s10208-011-9099-z.

[27] Anastasios A. Tsiatis. *Semiparametric Theory and Missing Data*. Springer Series in Statistics. Springer, 2006. doi: 10.1007/0-387-37345-4.

[28] Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer Series in Statistics. Springer, 2009. doi: 10.1007/b13794.

[29] Catherine Wah, Steve Branson, Peter Welinder, Pietro Perona, and Serge Belongie. The Caltech-UCSD Birds-200-2011 dataset. Technical Report CNS-TR-2011-001, California Institute of Technology, 2011. URL https://www.vision.caltech.edu/datasets/cub_200_2011/.

[30] Bolei Zhou, Agata Lapedriza, Aditya Khosla, Aude Oliva, and Antonio Torralba. Places: A 10 million image database for scene recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(6):1452–1464, 2018. doi: 10.1109/TPAMI.2017.2723009. URL https://doi.org/10.1109/TPAMI.2017.2723009.